Jet physics to probe the properties of Quark-Gluon Plasma (QGP), and development of level 1 trigger system in the EMCal/DCal detectors in ALICE

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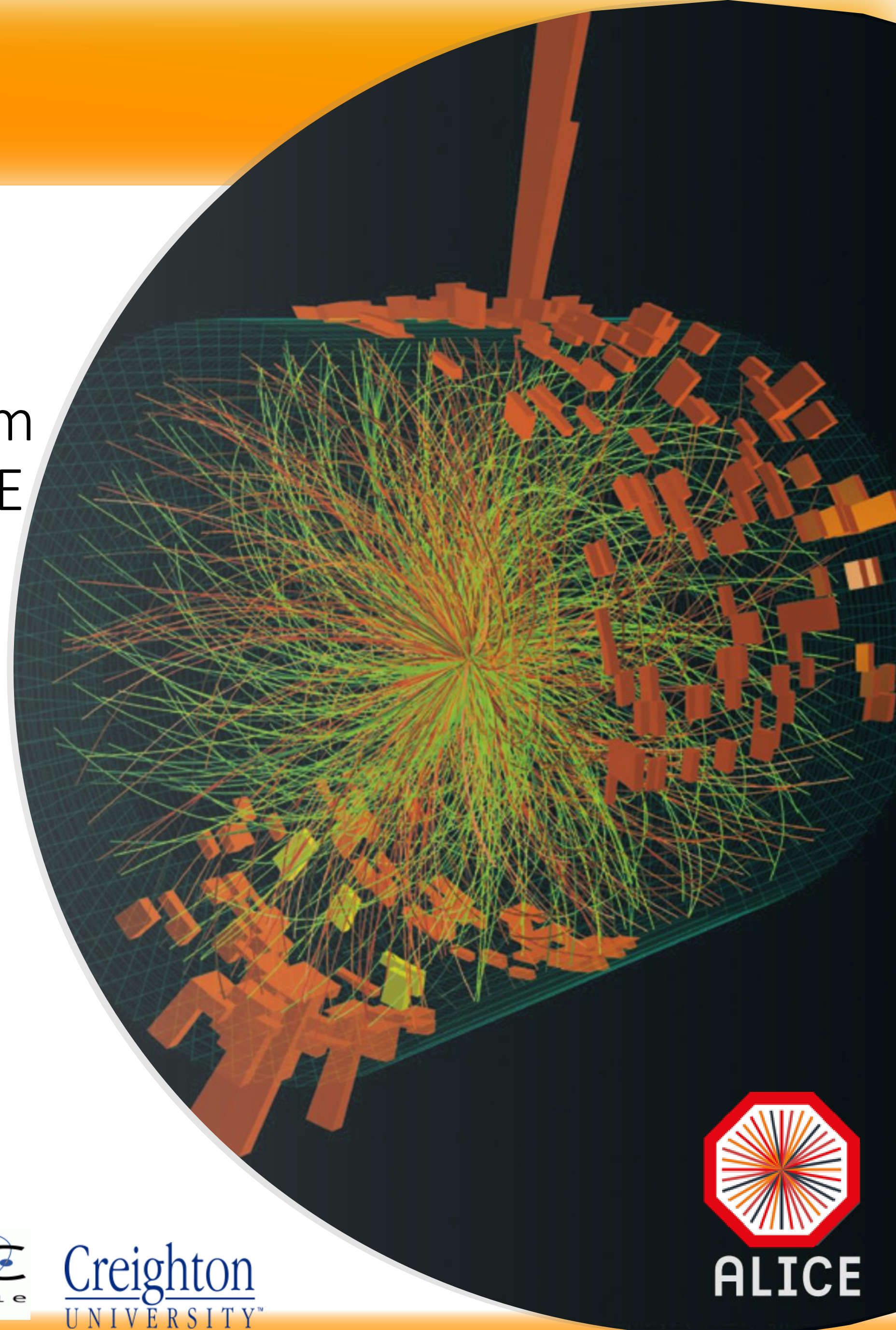


Outline

- Introduction
  - Quark-Gluon Plasma and jets
- Level-1 trigger system
  - Trigger system upgrade for the ALICE calorimeters
- Jet measurements
  - Charged jet production cross section in pp collisions at $\sqrt{s}$ = 5.02 TeV
  - Charged jets in mid-central Pb-Pb collisions at $\sqrt{s_{\mathrm{NN}}}$ = 5.02 TeV
    - Measurement of Jet v_2
    - Measurement of Jet-hadron correlation
- Summary

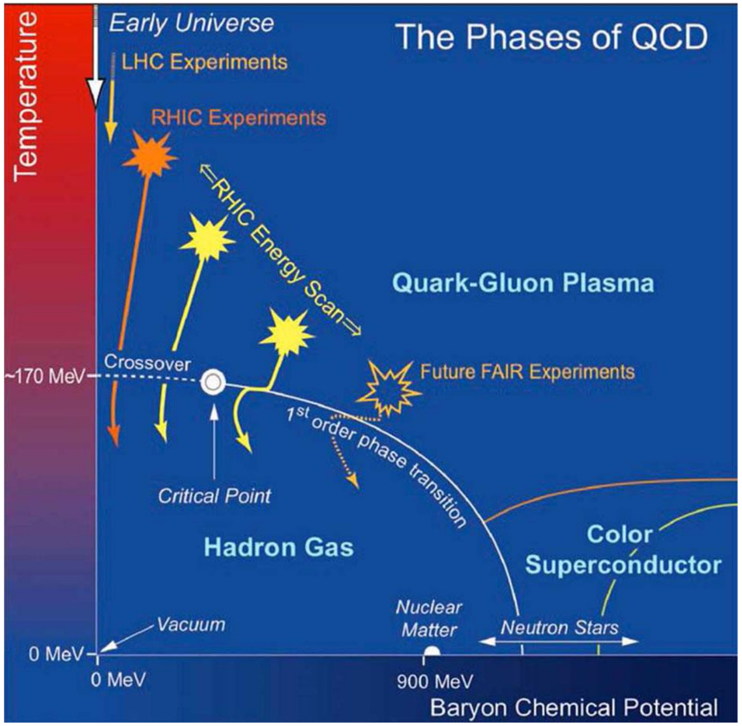
Finite temperature QCD

- Partons (quarks and gluons) are confined into hadrons under usual conditions
  - Partons cannot be isolated (quark confinement)

What happens under extreme conditions?
(Very high temperature, pressure…)

- Nucleons/mesons decouple and partons start to move over large volume
(~$10^0 – 10^2$ fm^3 in heavy-ion collisions)

<u>Quark-Gluon Plasma (QGP)</u>



Jet: powerful tool to probe the QGP properties

Jets

- Collimated sprays of high momentum particles which are originated from initial hard scattered partons
- In general, Jets are produced back-to-back (Di-Jet)

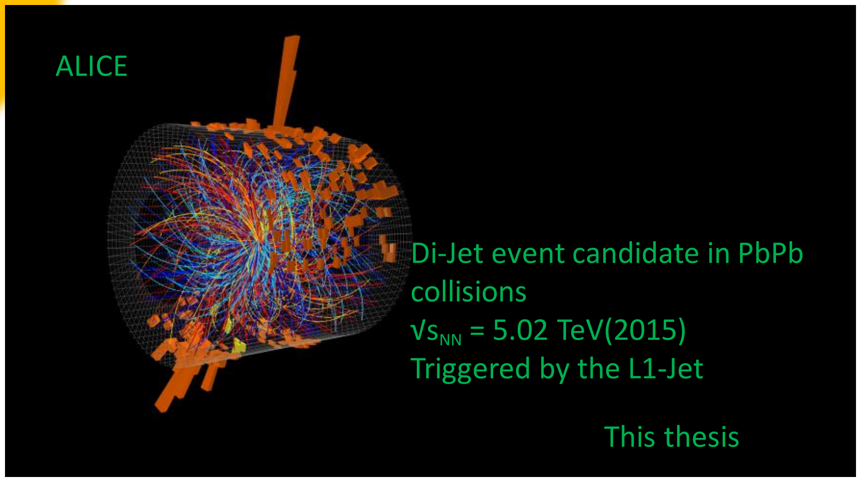
Remarkable feature

- The elementary process is well described theoretically, and production cross section is calculable with pQCD
- Self produced probes
  - The QGP life time is very short (~10^{-23} s)

In medium interaction

- Initial parton energy will be suppressed while partons are traversing the QGP and jet properties will be modified
  - Collisional energy loss ($\Delta E \propto L$)
  - Radiative energy loss ($\Delta E \propto L^2$)

➡ **Jet quenching**



The purpose and motivation of this study

Calorimeter trigger system upgrade

- DCal installation during LHC Long Shutdown 1 (LS1, 2013-2015)
  - ✓ Enhanced the acceptance
  - ✓ Improved the capability of jet measurements

Efficient event trigger is a powerful tool for measurements with high statistical precision

- **<u>Firmware upgrade and commissioning</u>**

✓ Adapted for new detector system configuration

✓ New trigger algorithm for high-p_T photon, electron and jet events

The purpose and motivation of this study

Data analysis

➢ Charged jet measurements in pp and Pb-Pb collisions

pp collisions

(Phys. Rev. D 100, 092004 (2019))

Production cross section

- ✓ Measured for 5 < $p_{\mathrm{T,jet}}$ < 100 GeV/c by ATLAS at $\sqrt{s}$ = 7 TeV
- ✓ Compared to LO pQCD-based predictions → Not well described

How about more higher-order calculation?

➢ **<u>Measurement of production cross section at $\sqrt{s}$ = 5.02 TeV</u>**

- ✓ Compared to some LO and a **NLO** pQCD-based predictions

Reference for Pb-Pb collisions

Pb-Pb collisions

Jet R_{AA} as a function of centrality at $\sqrt{s_{\mathrm{NN}}}$ = 5.02 TeV

(Ph.D thesis, H.Yokoyama, TYL-FJPPL Young Investigator Award 2017)

- ✓ Strength of jet suppression was estimated as a function of in-medium mean path-length of partons

Dependence of jet profiles w.r.t event plane at $\sqrt{s_{NN}}$ = 5.02 TeV

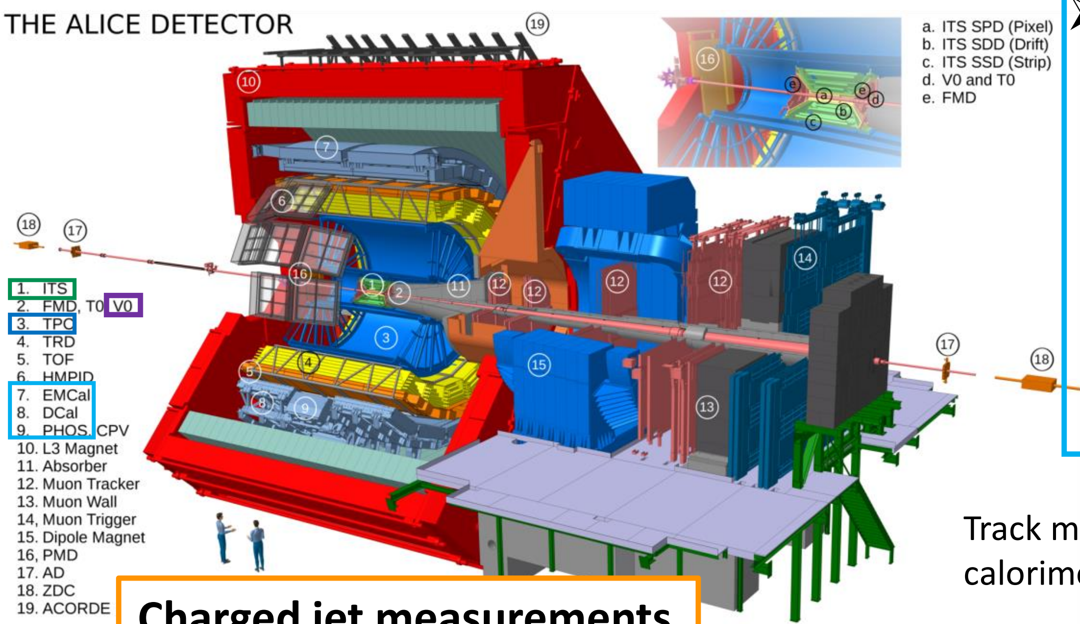
➢ **<u>Measurement of jet v_2</u>**

- ✓ Path-length dependence of parton energy suppression

➢ **<u>Measurement of jet shape through jet-hadron correlation</u>**

- ✓ Initial collision geometry dependence of suppressed energy re-distribution

Jet measurement at ALICE



Trigger system upgrade

- Calorimeters
  - EMCal, DCal
    - Sampling calorimeter with Pb-scintillator
    - Acceptance(EMCAL): $80^\circ < \phi < 187^\circ$, $|\eta|<0.7$
    - Acceptance(DCAL): $260^\circ < \phi < 327^\circ$, $|\eta|<0.7$
  - PHOS
    - Homogeneous calorimeter with PbWO4
    - Acceptance: $250^\circ < \phi < 320^\circ$, $|\eta|<0.12$

Remove contributions from charged particles

Track matching with calorimeter

Neutral constituents

Full Jets

Charged jet measurements

- Charged particle tracking
  - ITS (Inner tracking system)
    - Consists of three silicon detectors (SPD,SSD,SDD)
    - Acceptance: $0<\phi<2\pi$, $|\eta|<0.9$
  - TPC (Time projection chamber)
    - 3D tracking (pad position, drift time)
    - Acceptance: $0<\phi<2\pi$, $|\eta|<0.9$

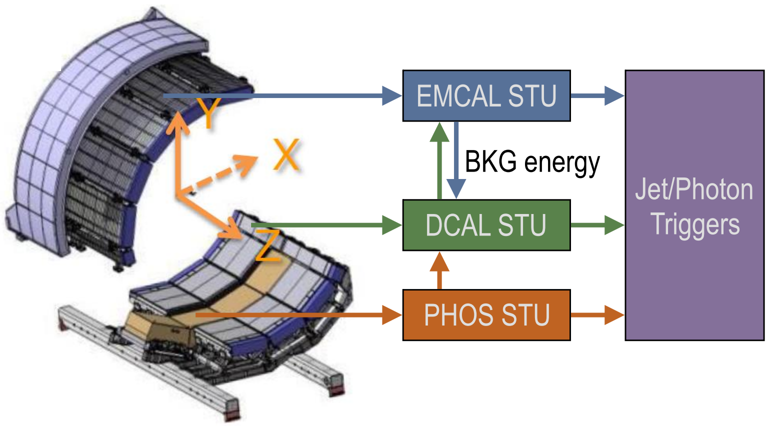
Charged constituents

Charged Jets

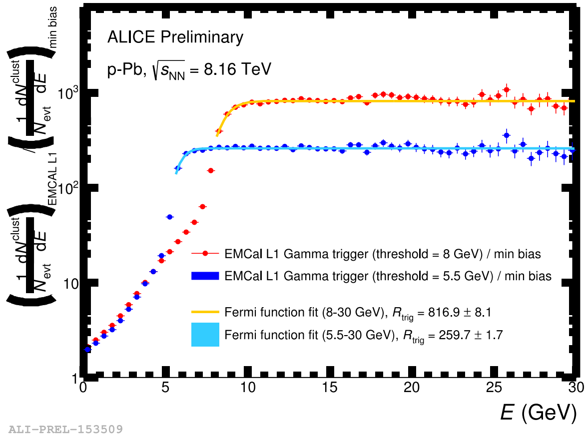
- V0
  - Multi-segmented scintillation counter
  - Acceptance: $0 < \phi < 2$, $2.8 < \eta < 5.1$ (V0A), $-3.7 < \eta < -1.7$(V0C)
  - Event trigger, Centrality, Event plane

ALICE calorimeter trigger upgrade

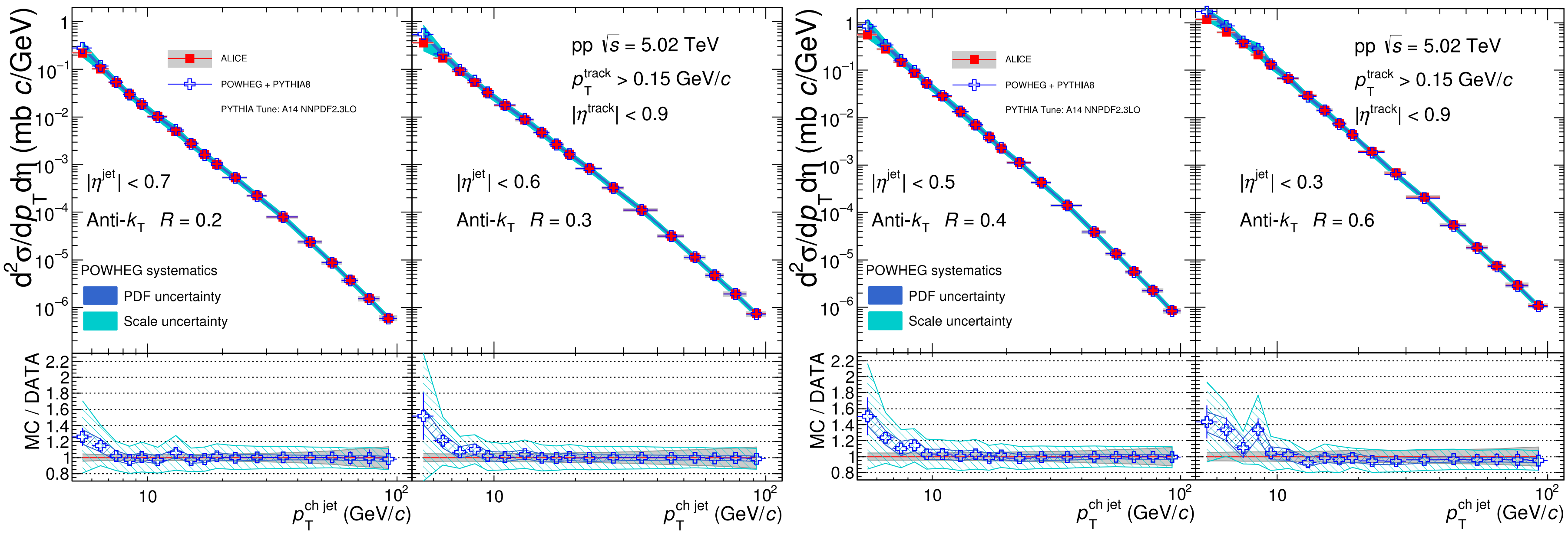
- Firmware development for STU
  - Upgrade of level-1 trigger algorithm
    - pp, p-Pb : constant threshold
    - Pb-Pb: Effective threshold variation event-by-event based on the background energy density estimated by opposite side calorimeter in azimuth
  - Adapted to new data stream convention and detector mapping
  - Upgrade of TRU-STU and TRU-TRU communication error monitoring system
  - DCS software upgrade
  - Commissioning tasks before 2015 Pb-Pb run



- Reasonable event rejection and turn-on curve
- Triggered data were successfully collected throughout LHC Run2 (2015-2018)
- New physics results are coming out with the L1 triggered data

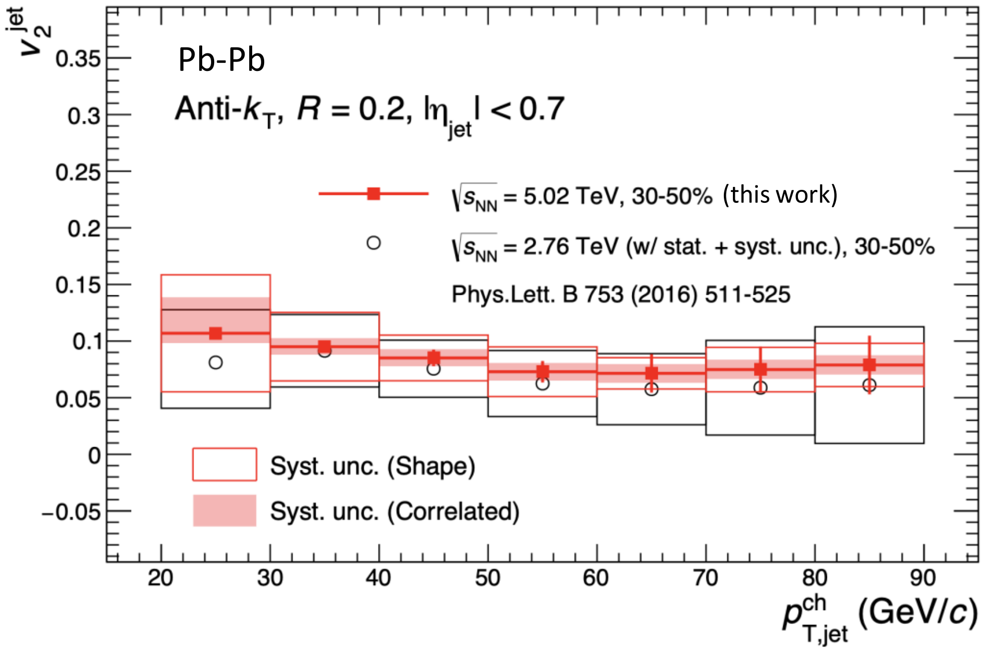


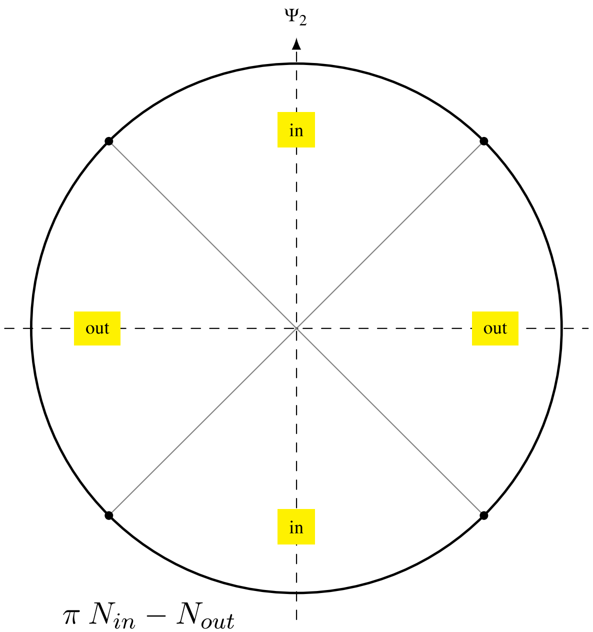
Charged jet cross section and comparison to Next-to-Leading-Order pQCD predictions in pp collsions



- Charged jet cross section in pp collisions at $\sqrt{s} = 5.02$ TeV is measured
- Comparison to a NLO pQCD-based model prediction (POWHEG+Pythia8)
  - Good agreement within large theoretical uncertainty
    - Higher-order (NNLO) calculation will improve scale uncertainties in pQCD calculation
    - Further understanding of non-perturbative effects (e.g. Underlying events) will also be crucial for low p_T region

Charged jet v_2 in Pb-Pb collisions at $\sqrt{s_{NN}}$ = 5.02 TeV





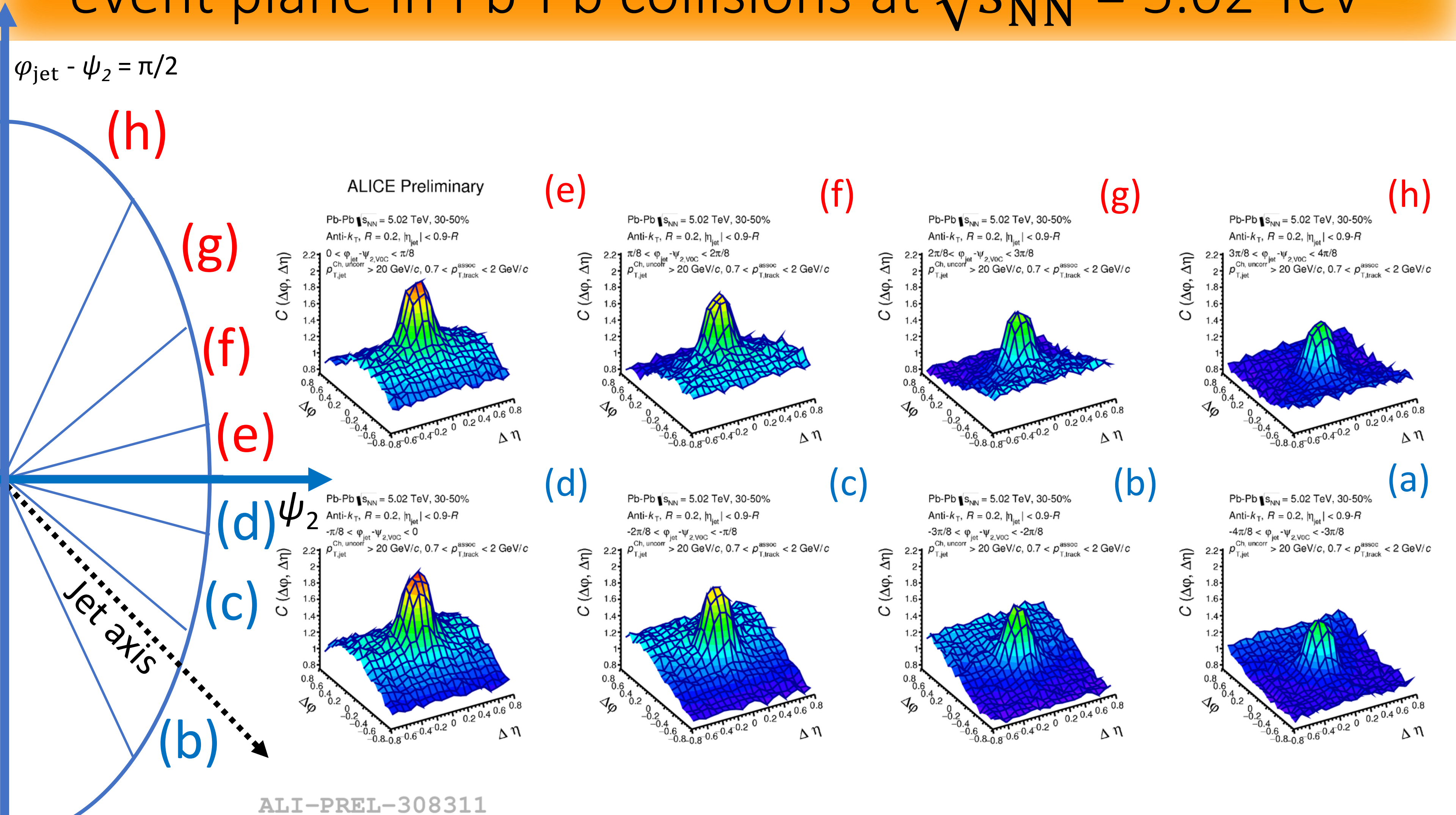
$$v_2^{jet} = \frac{1}{\mathrm{Res}\{\psi_2^{meas}\}} \frac{\pi}{4} \frac{N_{in} - N_{out}}{N_{in} + N_{out}}$$

N_{in}, N_{out} : Jet yield at in-plane and at out-of-plane

$\mathrm{Res}\{\psi_2^{meas}\}$: Event plane resolution

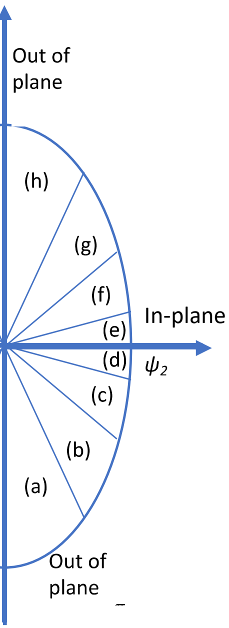
- **Positive jet v_2 was observed**
  - Jet yield in-plane is greater than that of out-of-plane
  - Can be interpreted as difference of in-medium parton path-length in-plane and out-of-plane
  - No significant collision energy dependence is observed

Correlation functions of the Near-side jet peak w.r.t event plane in Pb-Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

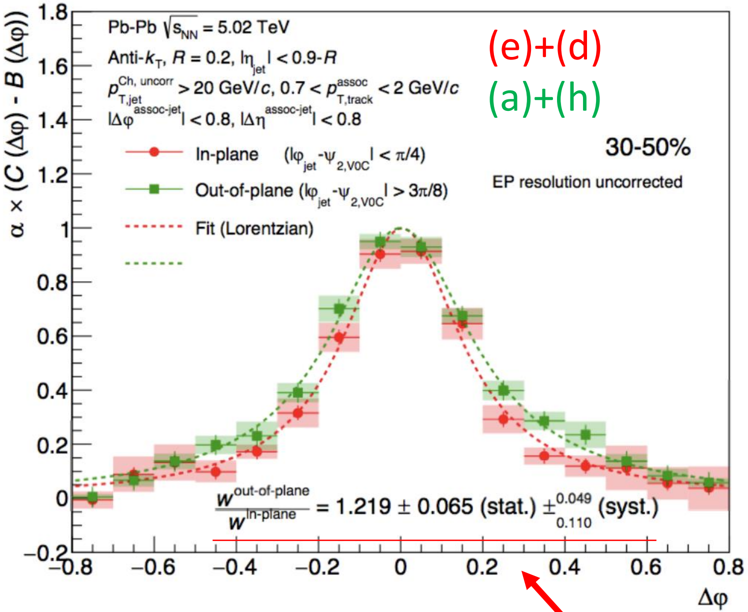


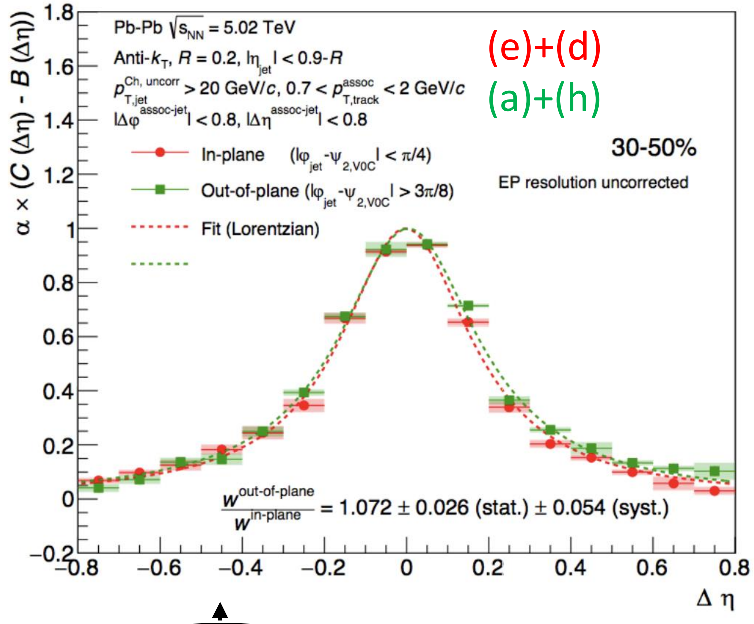
- Clear jet peak on top of the flow background
- Clear EP(Ψ_2) dependence of flow background
  - Flow background is estimated at side-band region of the correlation functions and subtracted

Jet-hadron correlation at in-plane and out-of-plane in Pb-Pb collisions at $\sqrt{s_{NN}}$ = 5.02 TeV

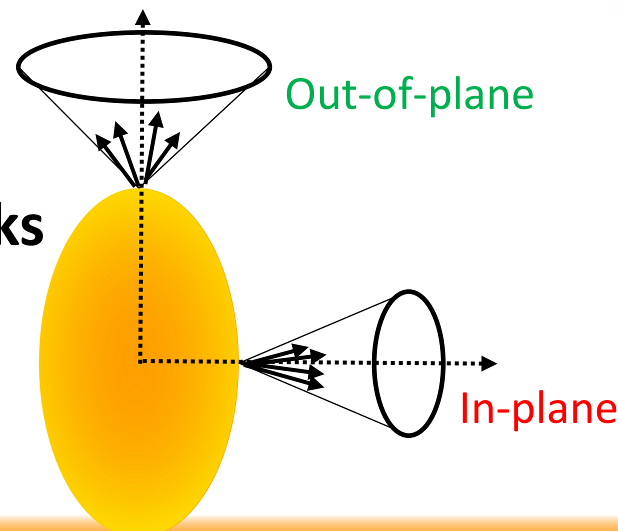


This work



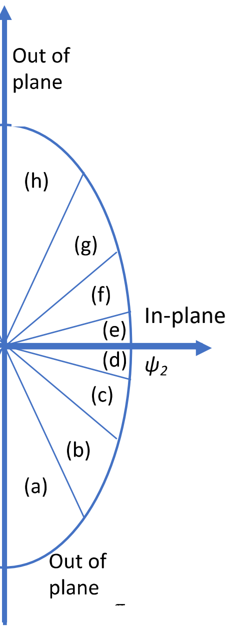


A slightly wider $\Delta\varphi$ distribution is observed out-of-plane for the lower p_T associated tracks ($0.7 < p_T < 2$ GeV/c)



- B($\Delta\varphi$), B($\Delta\eta$) : The background distributions that are dominated by flow
- α: scale factor chosen so that the peak height to be 1

Jet-hadron correlation at in-plane and out-of-plane in Pb-Pb collisions at $\sqrt{s_{\mathrm{NN}}} = 5.02$ TeV



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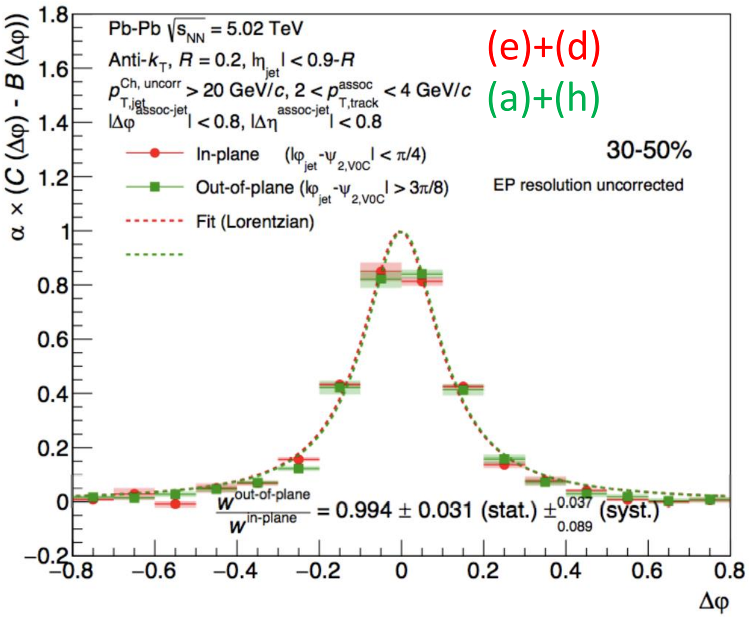
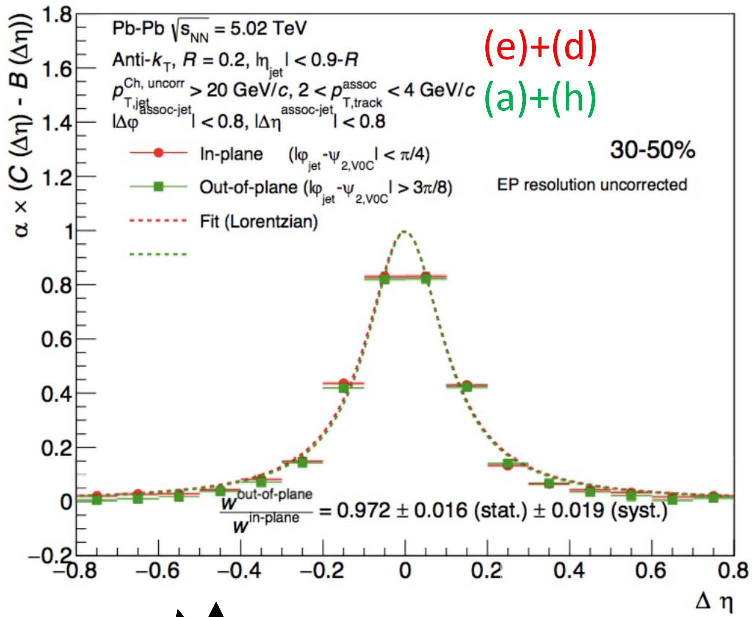
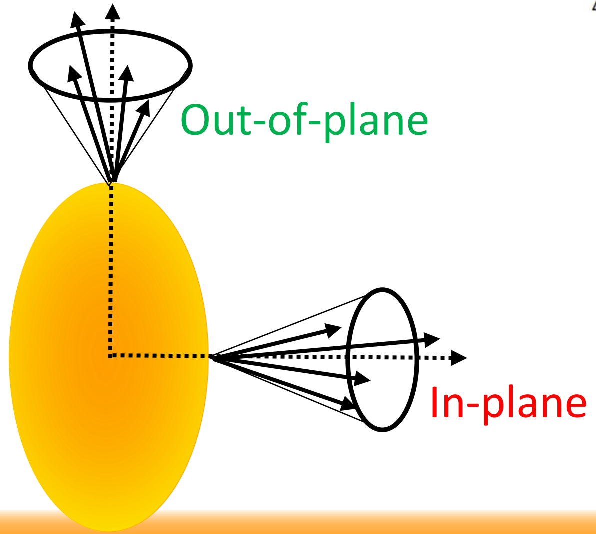




The broadening effect was not observed for higher p_{T} associated tracks ($2 < p_{\mathrm{T}} < 4$ GeV/c)



- **Jet broadening for lower p_{T} constituents**
- **No broadening effect for higher p_{T} constituents**

Summary: Level 1 trigger system in the EMCal/DCal detectors in ALICE

Trigger system upgrade

- ALICE calorimeter L1 trigger system was upgraded
  - New triggering algorithm
  - New detector configuration

- Triggered data were collected throughout LHC Run2 (2015-2018)
  - Data are enriched by hard probes
  - New physics results with high-statistics are coming out

Summary: Jet physics to probe the properties of Quark Gluon Plasma

Charged jet cross sections in pp collisions at $\sqrt{s}$ = 5.02 TeV

- LO,NLO pQCD calculations and models were tested
- High-statistics data took in 2017 will allow to study multiplicity dependent jet production

(Phys. Rev. D 100, 092004 (2019))

Charged jet v_2 in mid-central Pb-Pb collisions at $\sqrt{s_{NN}}$ = 5.02 TeV

- Positive charged jet v_2 was observed
- In-medium parton energy loss model will be constrained by reducing uncertainties and by combining with R_{AA} measurement
  - e.g.) High-statistics data took in 2018, UE fluctuation reduction with machine learning technique

Charged jet-hadron correlation of near-side jet peak in mid-central Pb-Pb collisions at $\sqrt{s_{NN}}$ = 5.02 TeV

- Jet shape modification w.r.t 2nd-order event plane was observed
- Centrality dependence will give more global picture of the geometry dependence

Thank you for your attention

Merci pour votre attention

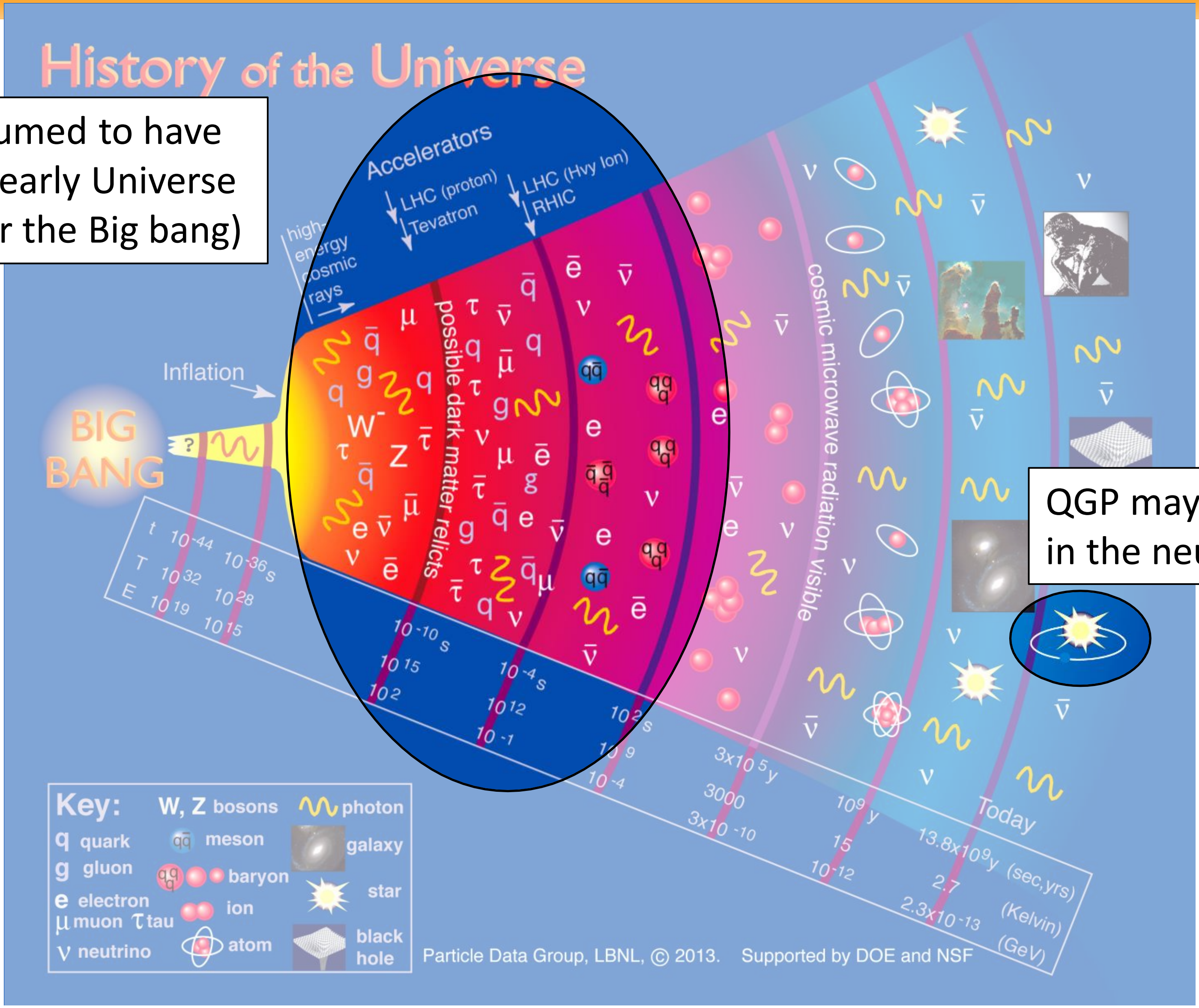
끝까지 경청해 주셔서 감사합니다

ご清聴ありがとうございました

Backup

QGP in the Universe

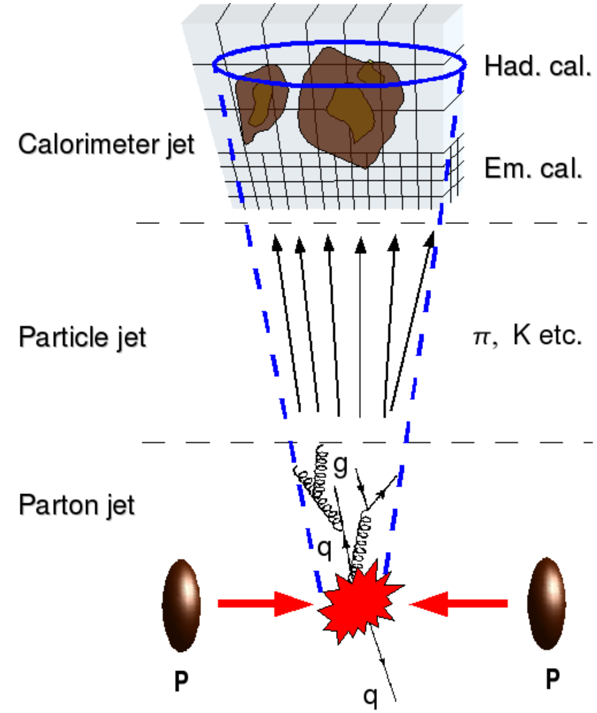
QGP is presumed to have formed the early Universe (~10 μs after the Big bang)



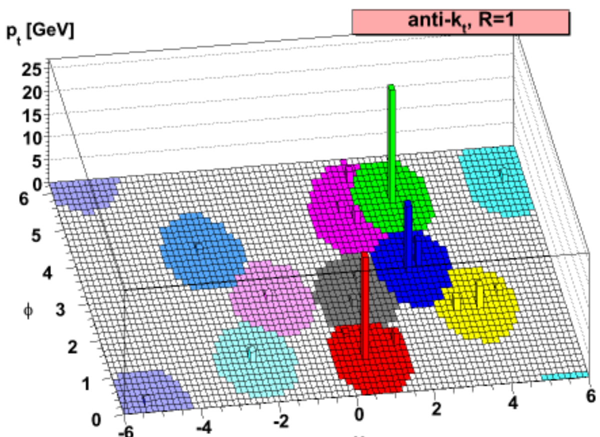
QGP may be formed in the neutron stars

Jet reconstruction

- Measure of the initial parton energy/momentum with final state particles
  - Jet reconstruction algorithms
    - Allow correspondence between detector level jet measurement and theoretical calculation



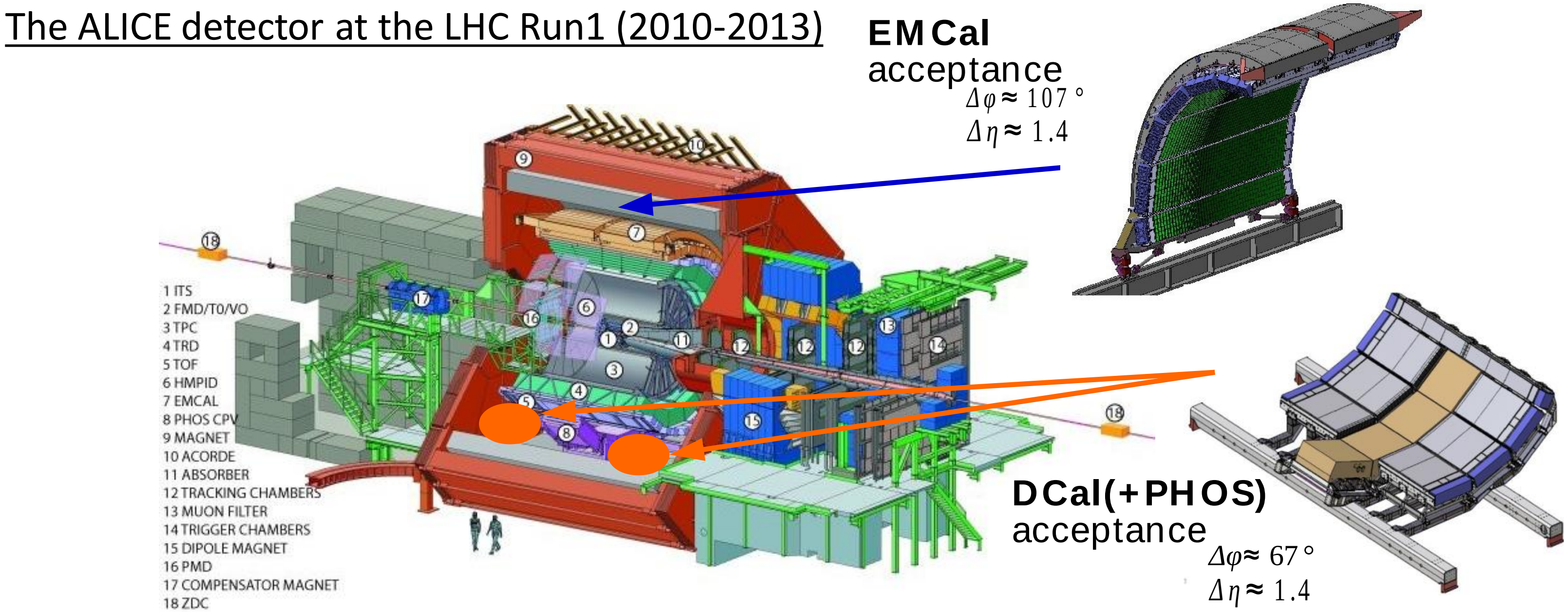
R: Jet resolution parameter (cone radius)



JHEP 0804:063,2008

- **'Anti-k_T algorithm'** is used for signal jets reconstruction
  - Clustering algorithm
  - Infra-red and collinear (IRC) safe
  - Circular cone shaped jet
  - Currently, one of the standard algorithms

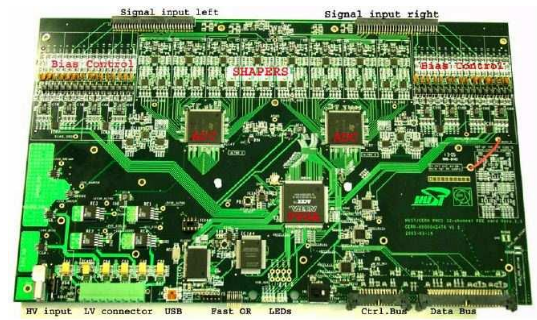
ALICE calorimeter upgrade during the LHC Long Shutdown 1



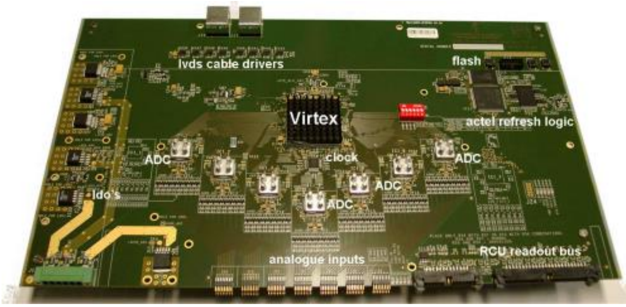
- Di-jet Calorimeter (DCAL) installation during LS1
  - back-to-back in azimuth to the ElectroMagnetic Calorimeter (EMCal)
  - **Enhanced jet measurement capability**
    - Acceptance is 44% increased
    - Improvement of Di-jet energy resolution
  - **Efficient trigger is crucial for high energy jet and photon measurement with high statistics**

Components of ALICE calorimeter trigger system

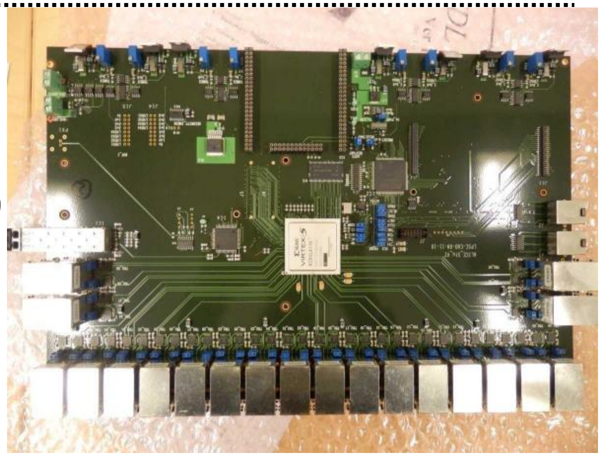
- FEE (Front End Electronics)
  - Takes 32 analog inputs and generates 4 towers/crystals analog sums (fastOR)



- TRU (Trigger Region Unit)
  - Samples and digitizes fastOR signals from FEE
  - Computes L0 (Level-0) trigger

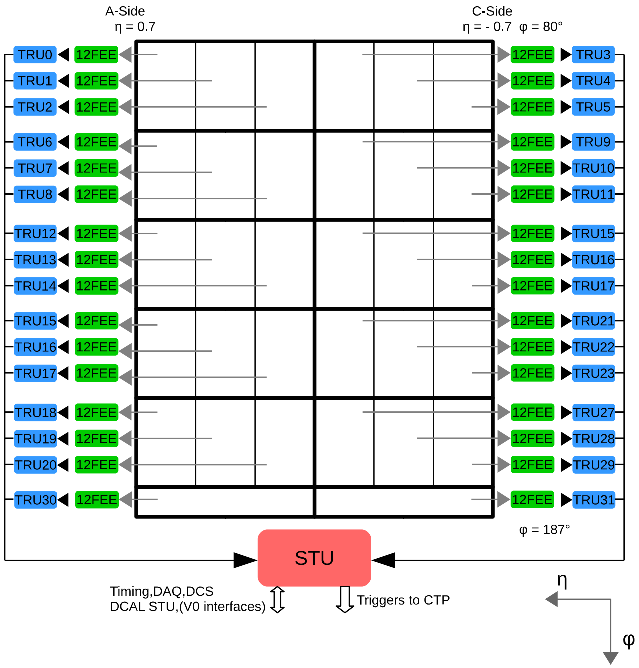


- STU (Summary Trigger Unit)
  - Computes L1 triggers
  - Transmits trigger inputs to CTP (Central Trigger Processor)
  - Transmits all data for trigger calculation from FEEs and TRUs to DAQ

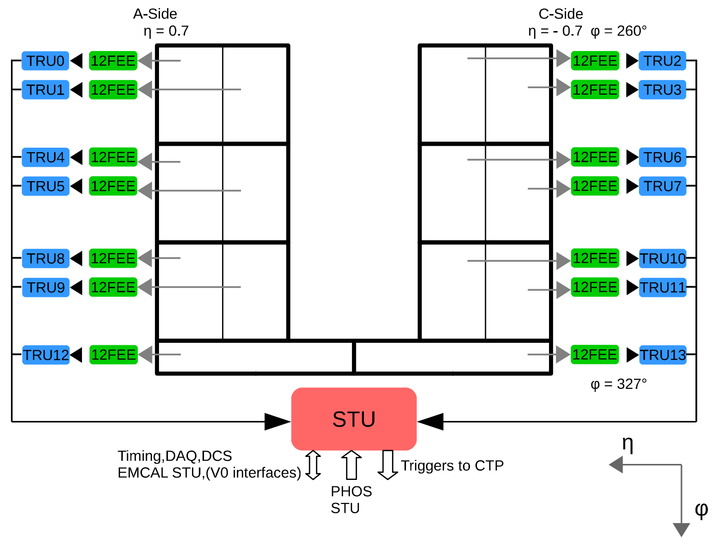


Components of ALICE calorimeter trigger system

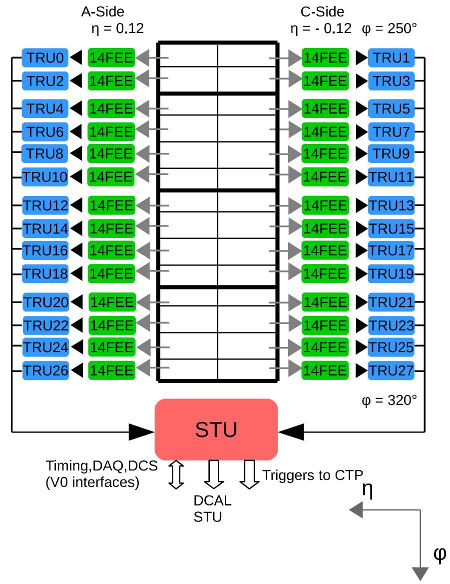
➢ EMCal



➢ DCal

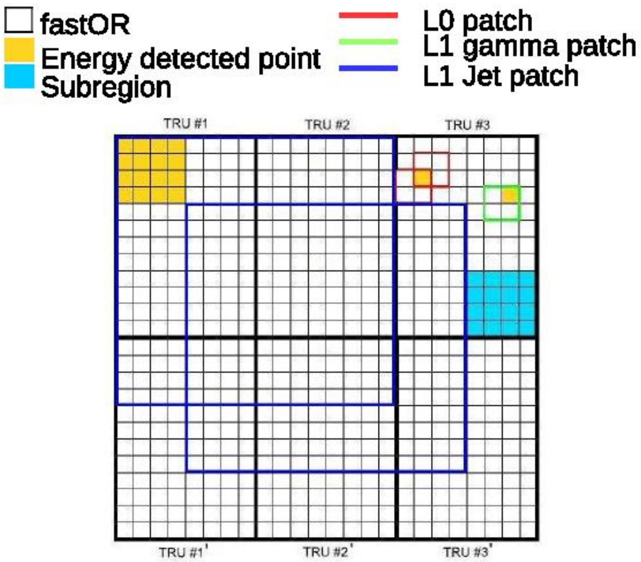


➢ PHOS

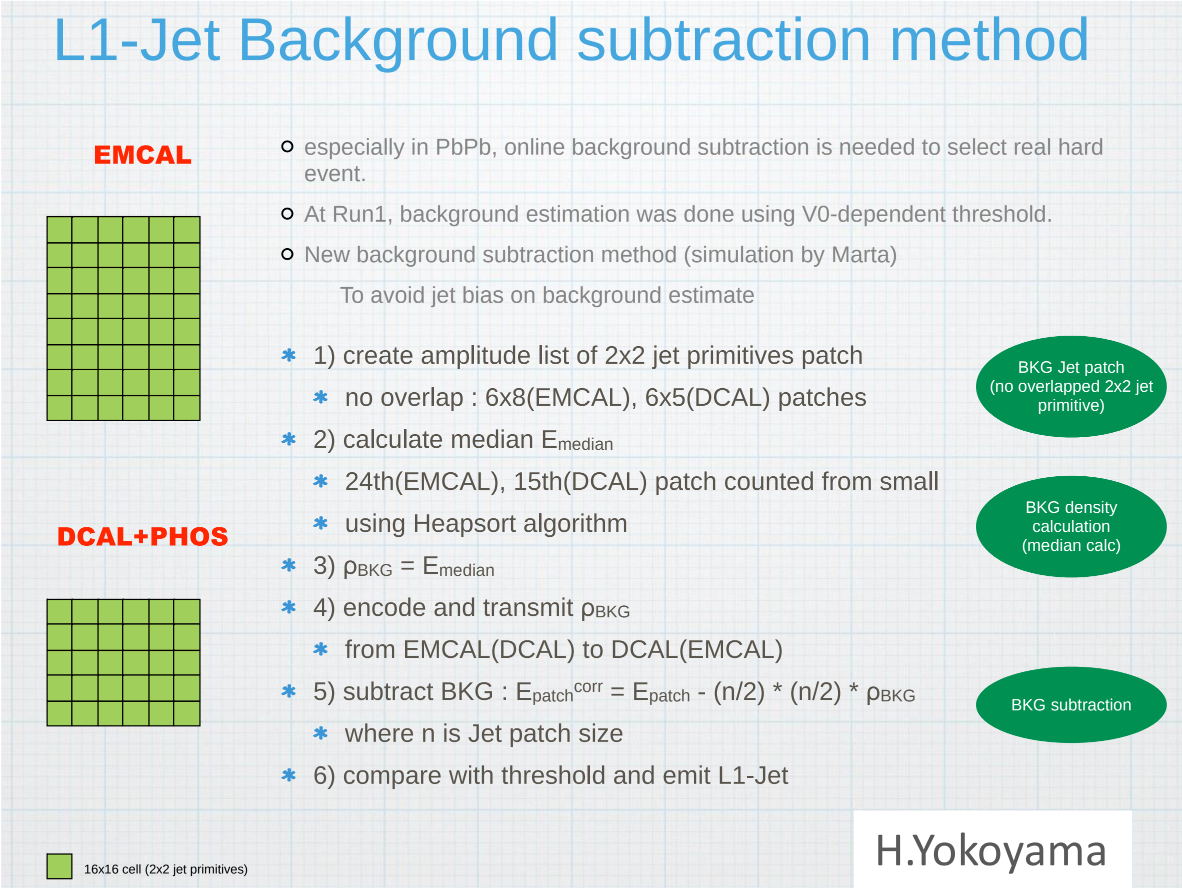


Triggering algorithm

- Patch algorithm



- Event-by-Event Background estimation (A-A, p-A)



k_T and Anti-k_T clustering algorithm

- Calculate d_{ij} and d_{iB} all particle combinations (p=1: *k*T , p=-1: Anti-k_T)

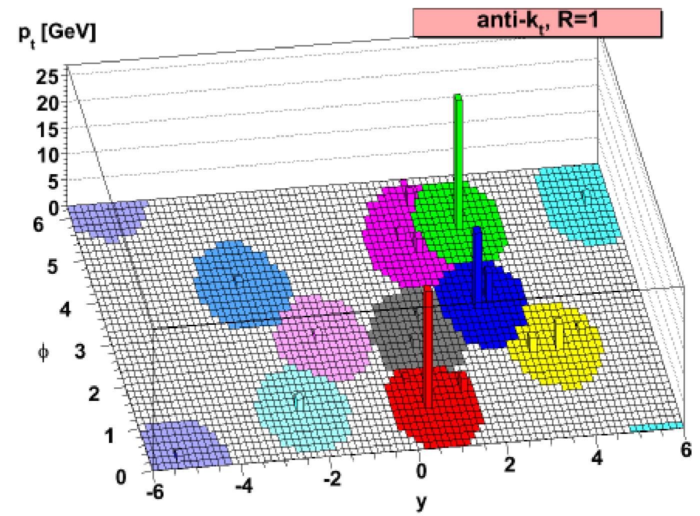
$$d_{ij} = \min(k_{ti}^{2p}, k_{tj}^{2p}) \frac{\Delta_{ij}^2}{R^2},$$

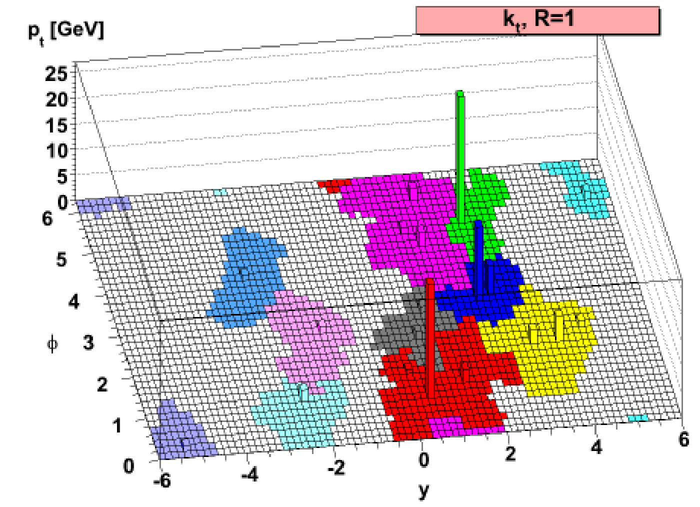
$$d_{iB} = k_{ti}^{2p},$$

  - Caluculate the minimum value of d_{ij} and d_{iB} as $d_{\min}$
    - $d_{min} \in d_{ij}$
      - Combine the particles i and j
    - $d_{min} \in d_{iB}$
      - Consider the cluster as a jet
- Repeat above procedure until all particles are clusterized

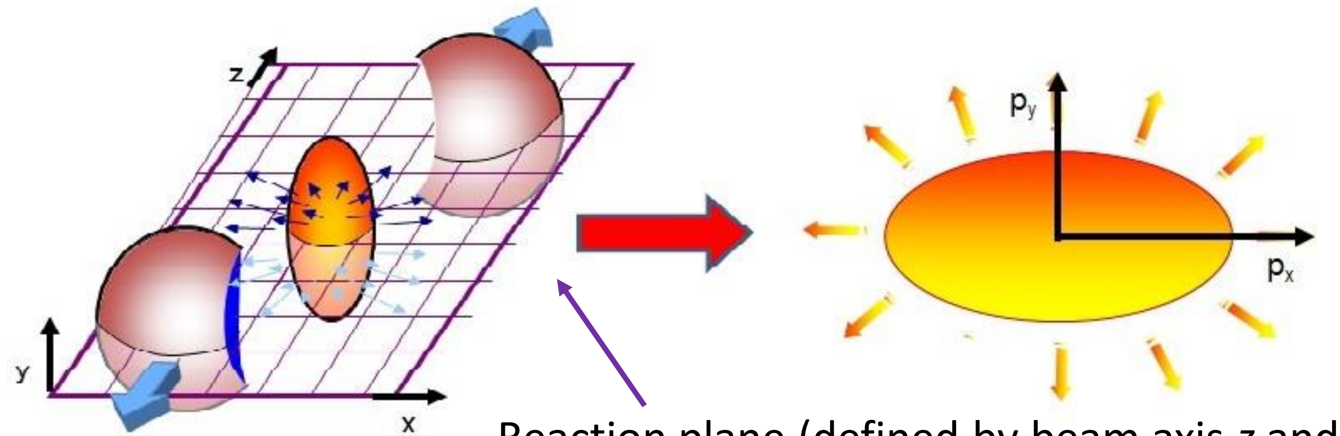
$$\Delta_{ij}^2 = (y_i - y_j)^2 + (\varphi_i - \varphi_j)^2$$

R: Jet resolution parameter
k: transverse momentum
y: (Pseudo) rapidity
φ: azimuth





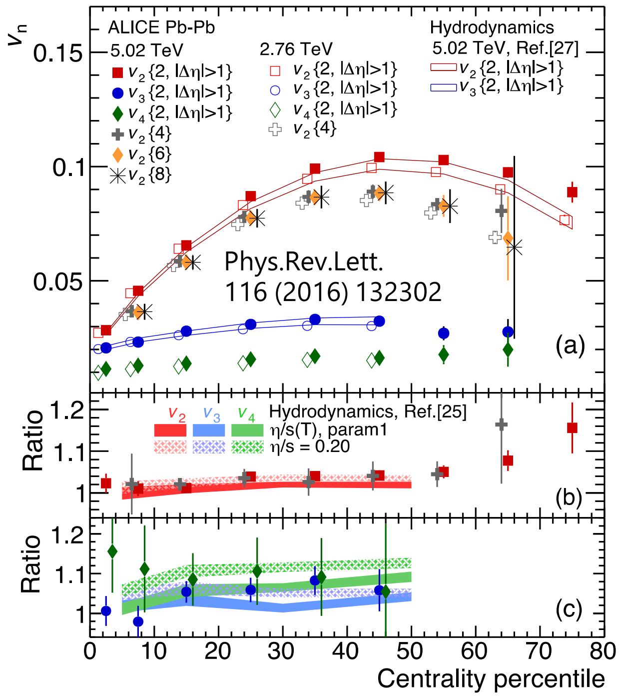
Elliptic flow in heavy-ion collisions

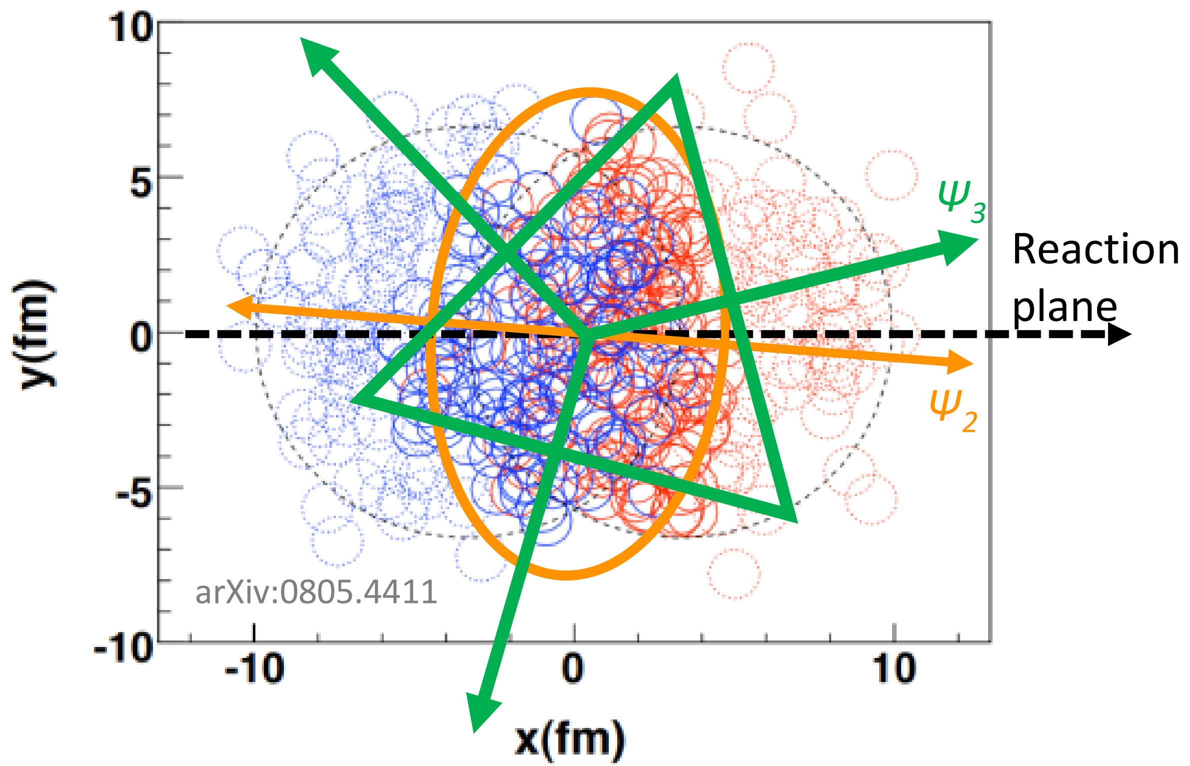


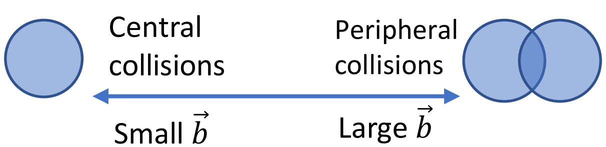
Reaction plane (defined by beam axis *z* and impact parameter $\vec{b}$)

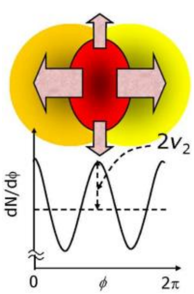
- In non-central collisions:
  - QGP will be formed in the almond shaped overlap region
  - Pressure gradient differ in-plane and out-of-plane
    - Azimuthal anisotropy of hadron production in momentum space
      - Smaller $\vec{b}$ (most central → mid-central collisions) → large overlap region eccentricity → large anisotropy
- This process is the main background source for jet-hadron measurements

Azimuthal anisotropy (v_n) and event plane (EP, ψ_n)









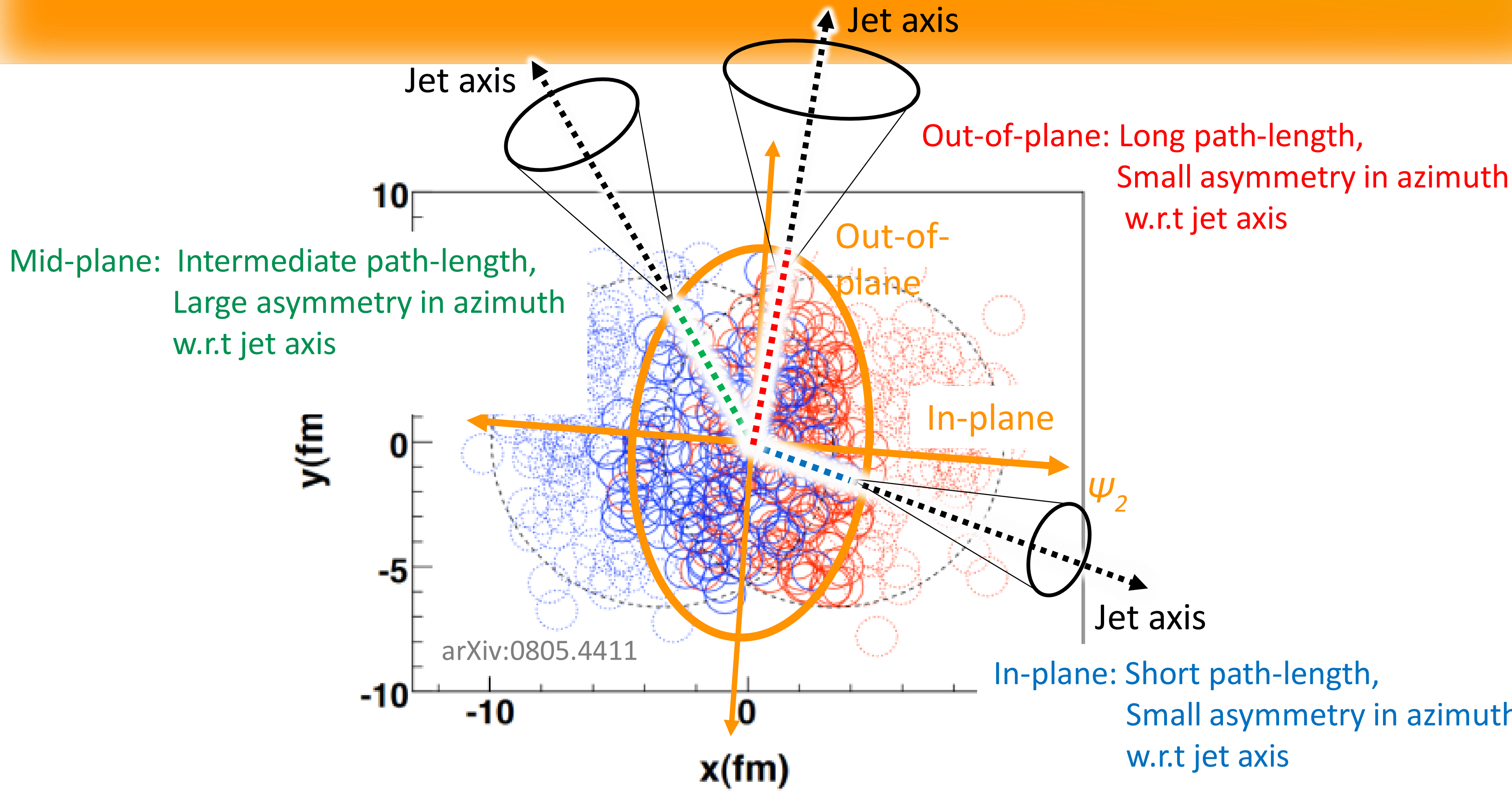
Azimuthal particle distribution: $$\frac{dN}{d\varphi} \propto 1 + \sum_{n=1}^{\infty} 2v_n \cos\{n(\varphi - \Psi_n)\}$$

- ➢ EP: Experimental access to initial collision geometry

EP for n-th order harmonics: $$\Psi_n = \frac{1}{n} \cdot \tan^{-1} \frac{\Sigma w_i \sin(n\varphi_i)}{\Sigma w_i \cos(n\varphi_i)}$$

w: weight, φ : Azimuthal angle of emitted particles

Initial geometry and parton path-length control

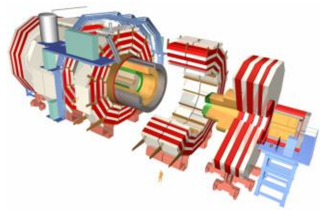


- Initial collision geometry and in-medium parton path-length can be constrained by selecting jet axis w.r.t. event plane

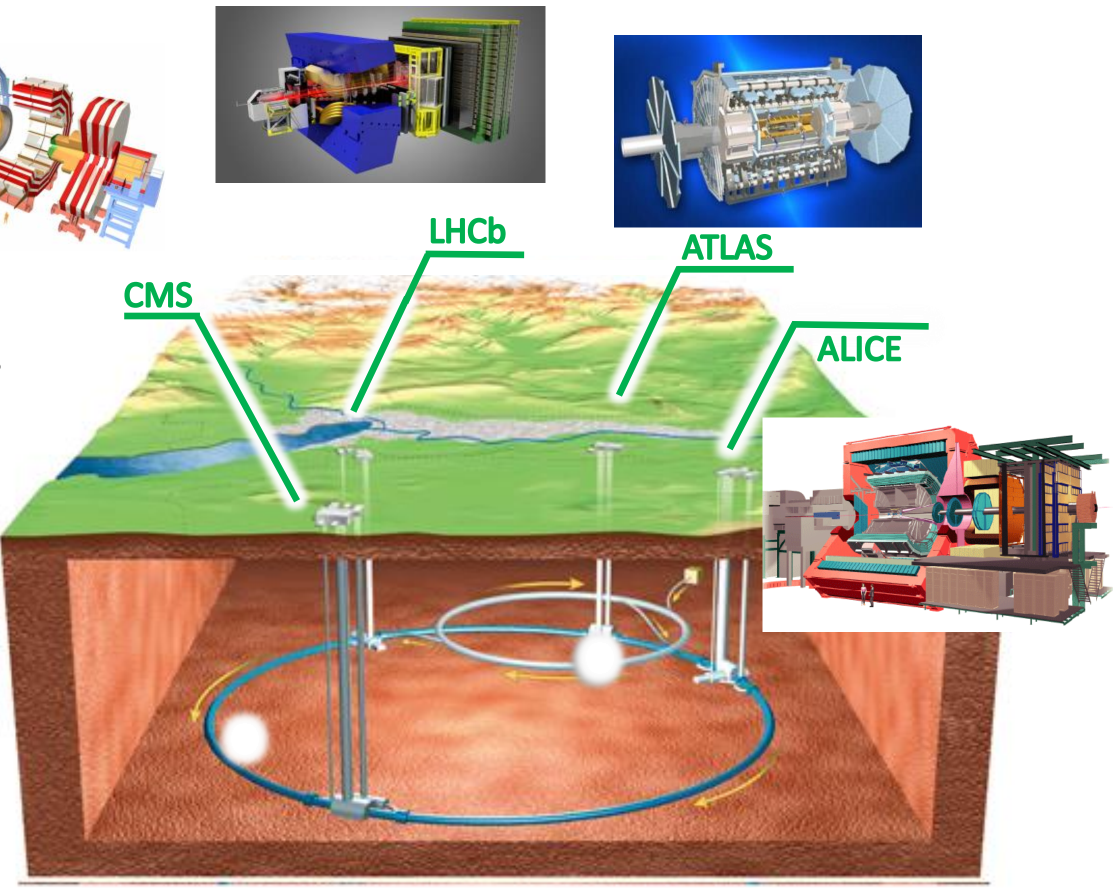
⇨ Initial collision geometry dependence of jet modification

The Large Hadron Collider (LHC)

- Currently, The world's largest particle accelerator built by CERN



- Particle species and collision energies
  - pp: $\sqrt{s}$ = 0.9, 2.76, 5.02, 7, 8, 13 TeV
  - Pb-Pb: $\sqrt{s_{NN}}$ = 2.76, **<u>5.02 TeV</u>**
  - p-Pb: $\sqrt{s_{NN}}$ = 5.02, 8.16 TeV

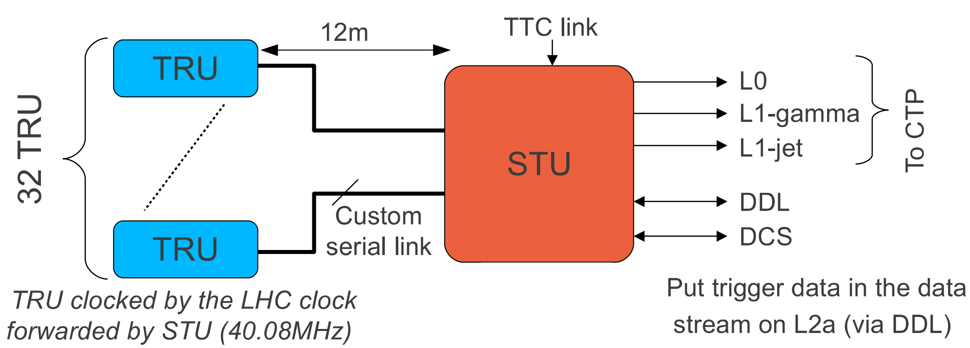


- Four major experiments
  - **<u>ALICE</u>**, ATLAS, CMS, LHCb

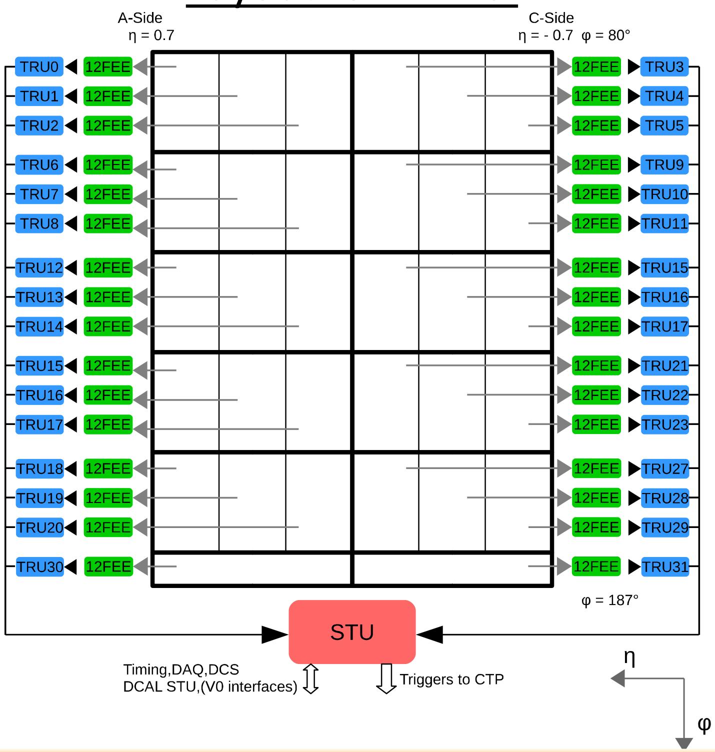
ALICE calorimeter trigger overview

- ALICE calorimeters (EMCal, DCal, PHOS) contributes both Level-0 (L0) and Level-1 (L1) triggers
  - L0: Fast trigger, Low threshold, Ensure hits on given detector
  - L1:
    - EMCal, DCal: High-energy photon, electron and jet trigger
    - PHOS: High-energy photon and electron trigger

- L1 trigger for jet events (L1-jet) and for photon, electron events (L1-gamma) are computed by Summary Trigger Unit (STU)

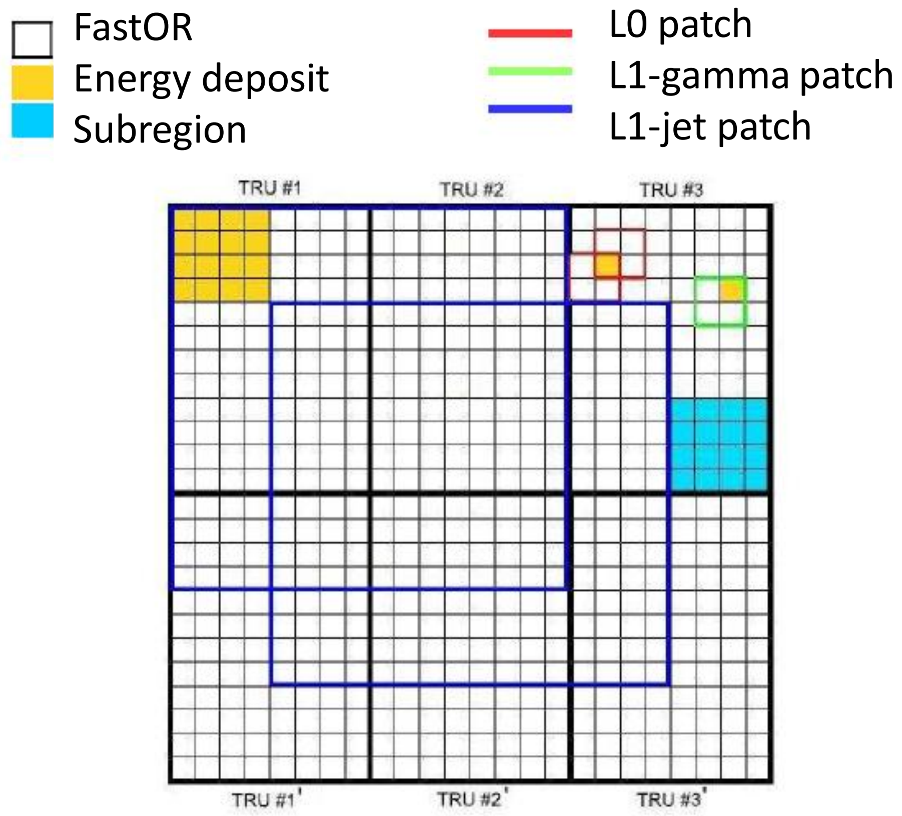


Layout for EMCal



Sliding window algorithm

- Trigger is issued if the sum of energy deposit within a patch exceeds threshold
  - L1-gamma patch
    - 2x2 FastOR
  - L1-jet patch
    - 8x8 or 16x16 FastOR



- FastOR:
  Corresponds to one 2x2 calorimeter cells

Analysis flow

Event selection

- pp collisions at $\sqrt{s}$ = 5.02 TeV
  - About 100 M minimum bias events (2 nb^{-1})

Track selection

- Charged tracks reconstructed by ITS + TPC
  - $|\eta_{track}| < 0.9$, $0.15 < p_{T,track} < 100$ GeV/c

Jet reconstruction

- Anti-kt clustering algorithm
  - Jet cone radii R = 0.2, 0.3, 0.4, 0.6
  - $|\eta_{jet}| < 0.9 - R$

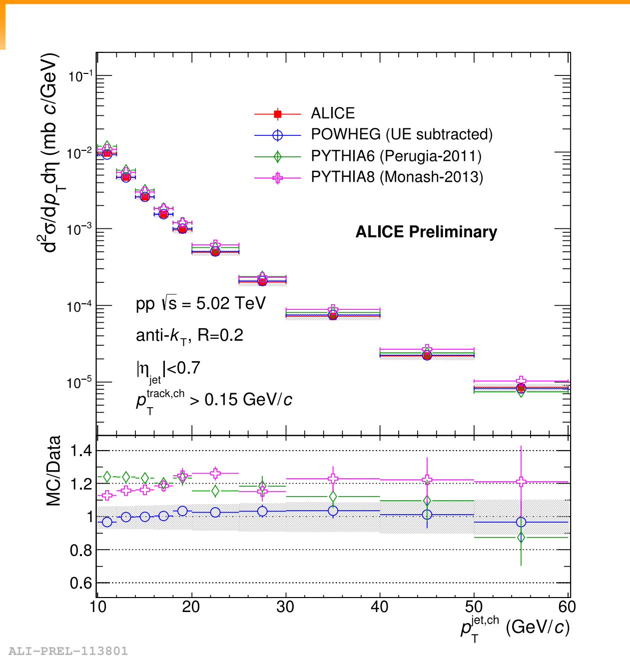
Correction for detector effects

- Corrected by an unfolding method (SVD)
  - Detector responses are extracted from full detector simulation with PYTHIA8 + GEANT3

Final Jet Spectrum

- Cross section: $\frac{\mathrm{d}^2\sigma_{\mathrm{jet}}}{\mathrm{d}p_T\mathrm{d}\eta} = \frac{\sigma_{pp}}{N_{\mathrm{trig}}^{\mathrm{evt}}} \cdot \frac{\Delta N^{\mathrm{jet}}}{\Delta p_T \Delta\eta}$

Basic concept of Unfolding



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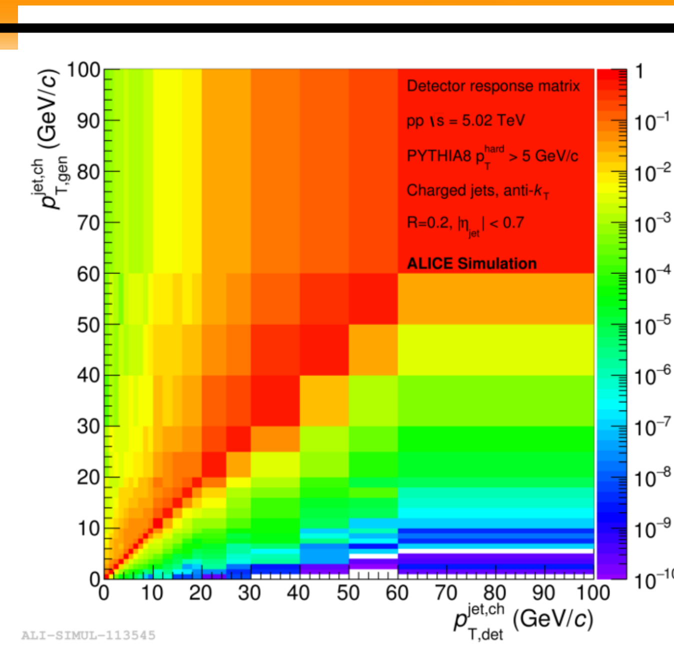

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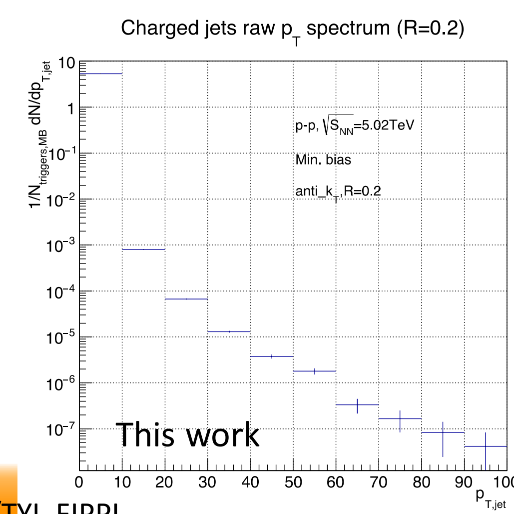


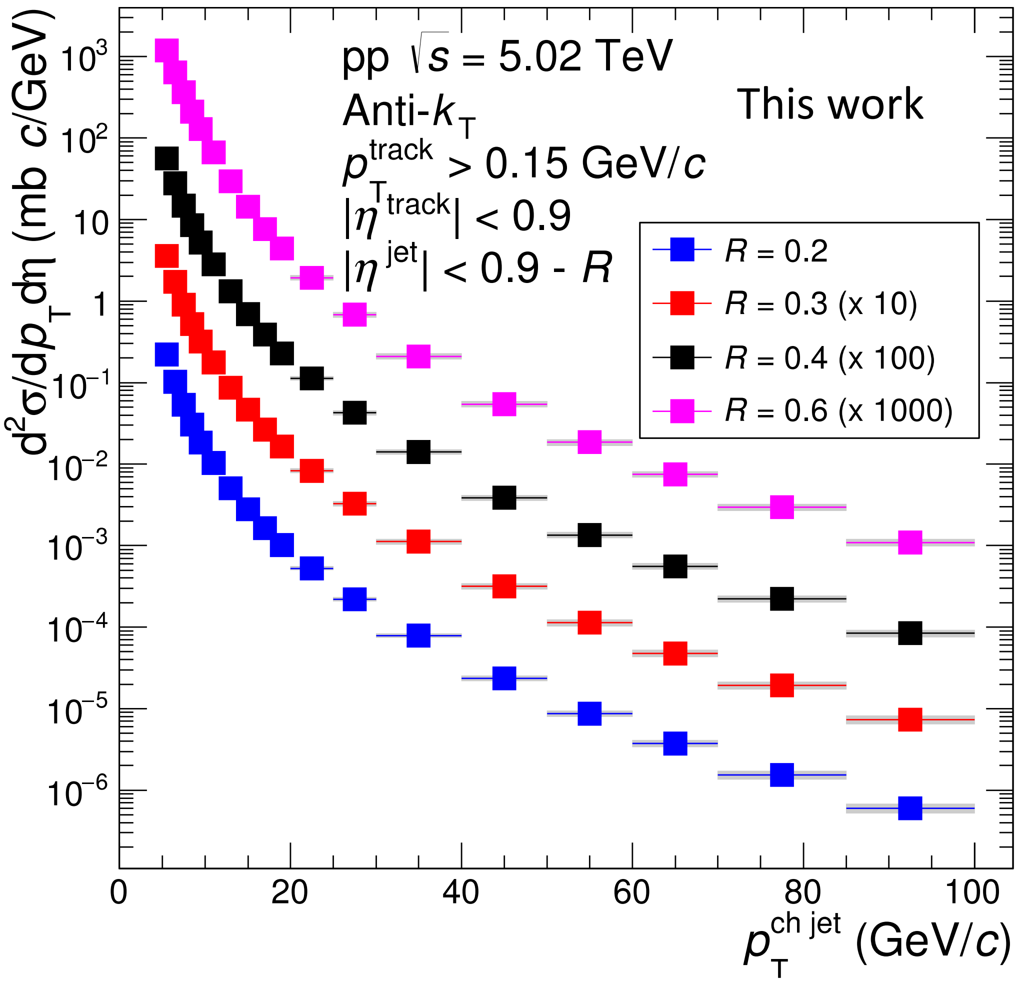
$$\mathbf{M}_m = \mathbf{R}^{\mathrm{det}}_{m,t} \cdot \mathbf{T}_t$$

Measured spectrum

Response matrix

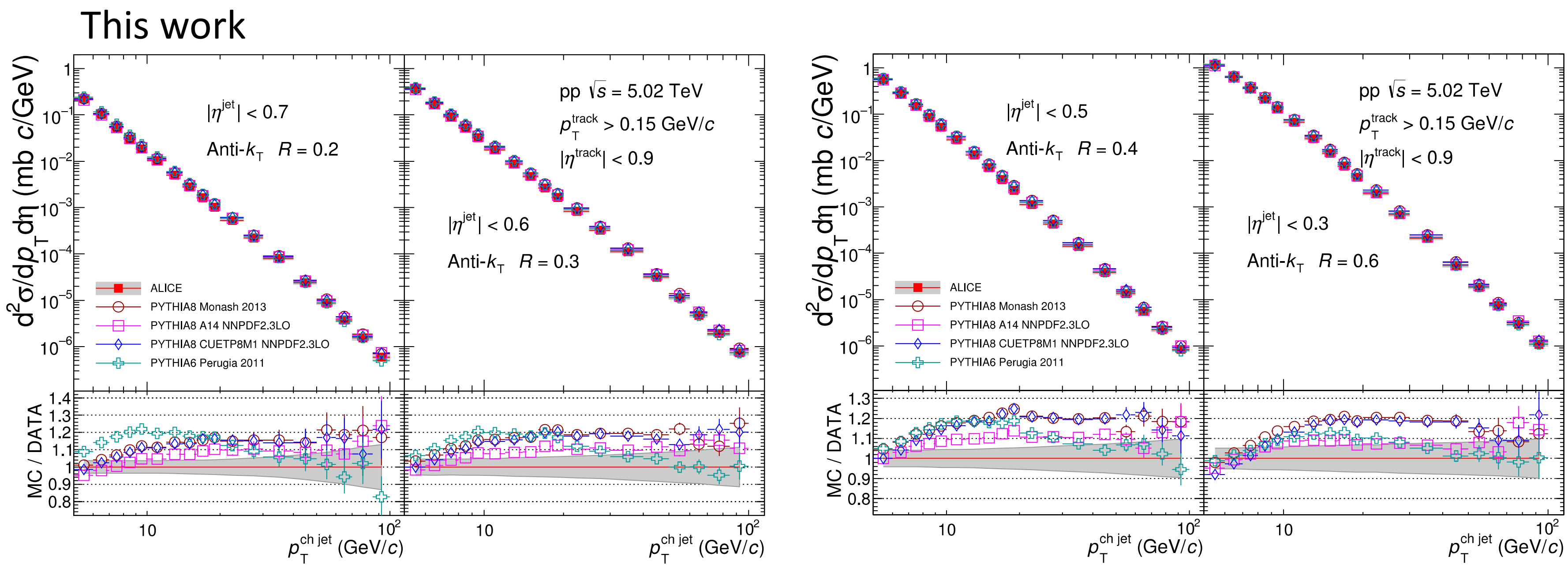
Unknown true spectrum

Charged jet production cross section



- Charged jet production cross sections with R = 0.2, 0.3, 0.4 and 0.6 were measured for 5 < $p_{T,jet}$ < 100 GeV/c

Comparison to Leading-Order(LO) pQCD predictions

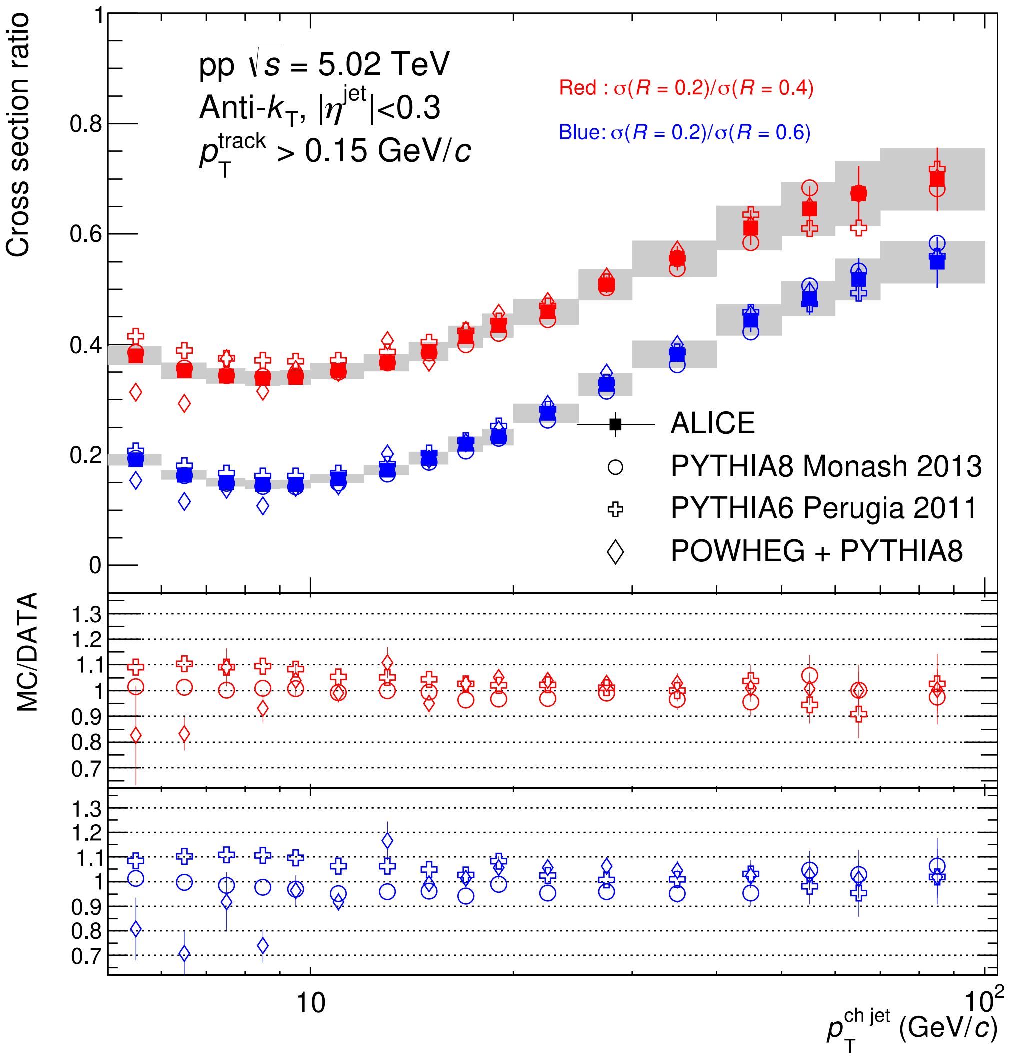


- ➢Comparison to several LO pQCD-based model predictions (Pythia8 with several tunes, Pythia6)
- ➢The spectra shape was not reproduced up to ~20 GeV/c
- ➢All predictions overestimate the charged jet production cross section at high-p_T (p_T > 10 GeV/c)

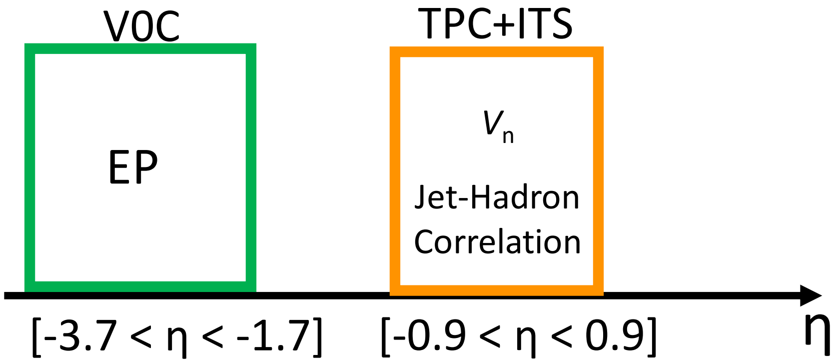
Charged jet cross section ratio

- Sensitive to jet radial profile
- All model predictions show good agreement at wide kinematic range
  ($10 < p_T < 100$ GeV/c)
  - Slight tension at the low p_T
    ($p_T < 10$ GeV/c)

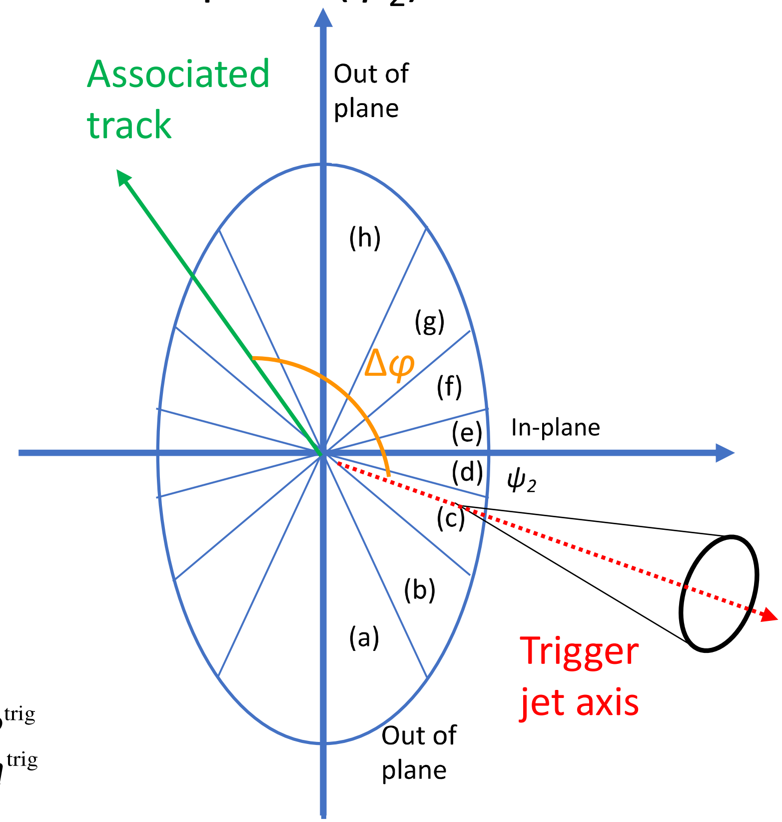
This work



Setup for jet-hadron correlation



- Jet events are classified w.r.t event plane (ψ_2)



- Trigger jets: $p_{\mathrm{T,jet}}^{\mathrm{Ch}} > 20$ (GeV/c)
  - Jet axes are recalculated with constituent tracks which have $p_{\mathrm{T}} > 4$ GeV/c to avoid auto correlation

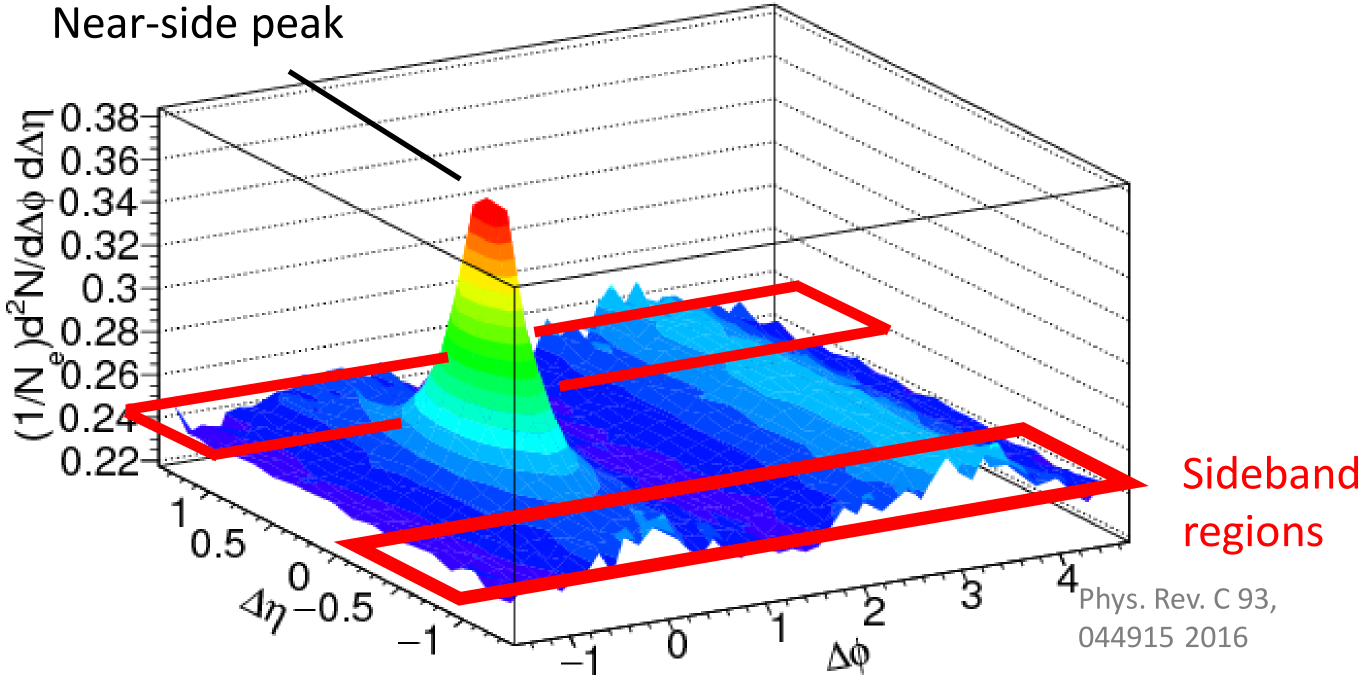
- Associate track p_{T} range: [0.7-2 (GeV/c)], [2-4 (GeV/c)]

- Correlation function

$$C(\Delta\varphi,\Delta\eta) = \frac{N_{\mathrm{mix}}^{\mathrm{pair}}}{N_{\mathrm{real}}^{\mathrm{pair}}}\left(\frac{d^2N_{\mathrm{real}}}{d\Delta\eta d\Delta\varphi} \Big/ \frac{d^2N_{\mathrm{mix}}}{d\Delta\eta d\Delta\varphi}\right) \qquad \begin{array}{l}\Delta\varphi = \varphi^{\mathrm{assoc}} - \varphi^{\mathrm{trig}} \\ \Delta\eta = \eta^{\mathrm{assoc}} - \eta^{\mathrm{trig}}\end{array}$$

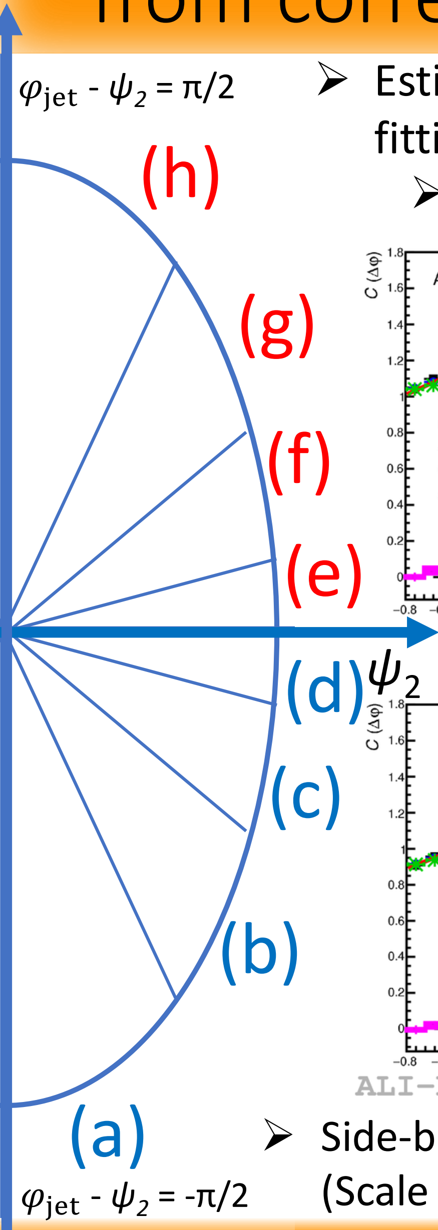
N_{real} , N_{mix} : Associated track yield in real and mixed event
$N_{\mathrm{real}}^{\mathrm{pair}}$, $N_{\mathrm{mix}}^{\mathrm{pair}}$: Total jet-track pairs in real and mixed event
φ , η : Azimuthal angle and pseudo rapidity of jet and tracks

Flow background subtraction from correlation function

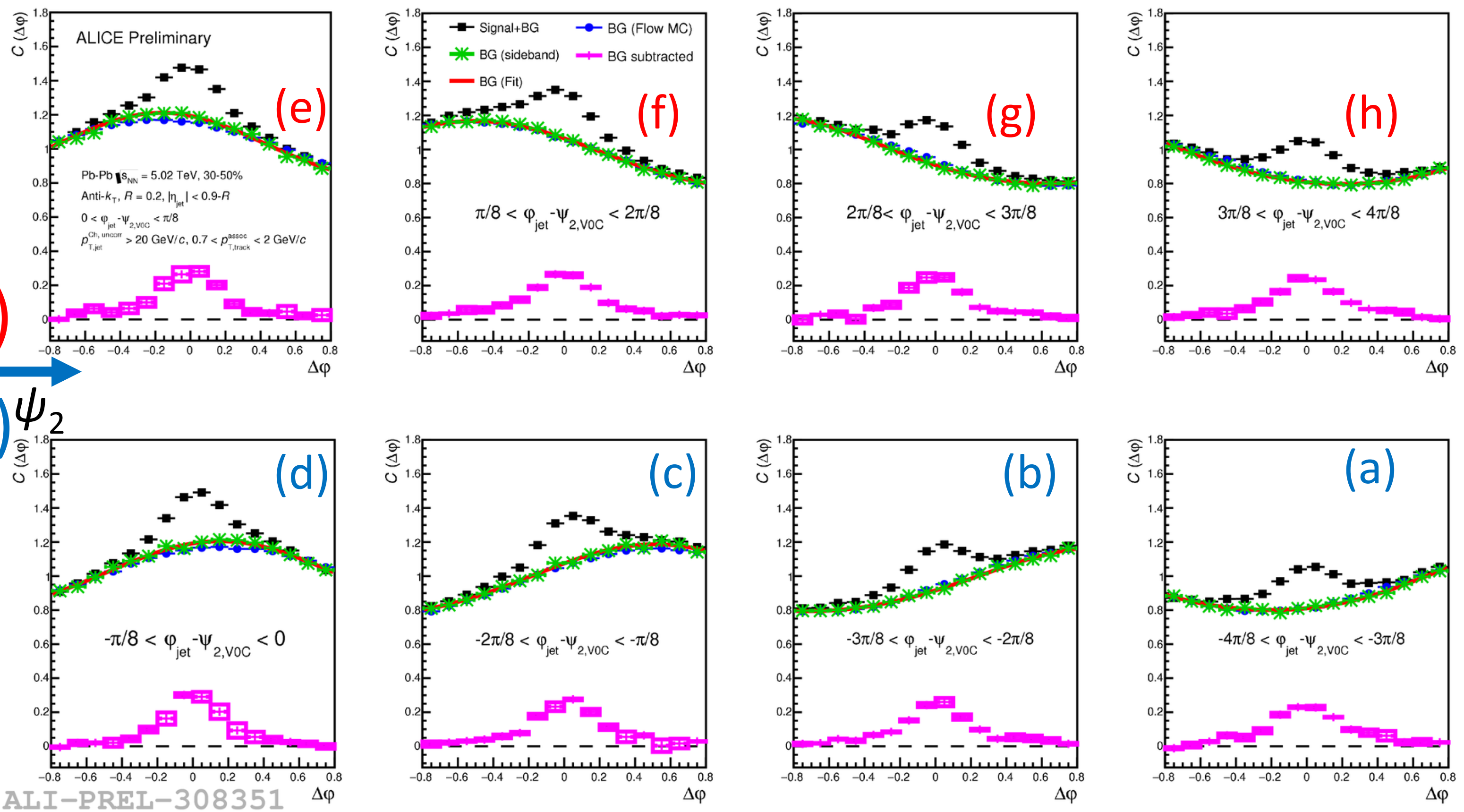


- The flow background is estimated at the sideband regions ($1<|\Delta\eta|<1.5$) of the correlation function
  - Several background estimation methods were tested as a systematic study
    - Estimation at the Narrow-side-band regions ($1.1 <|\Delta\eta|<1.3$)
    - Fitting to side-band region distributions
    - Flow Monte-Carlo simulation
  - The background estimation methods is the largest systematic uncertainty source for this measurement

Flow background subtraction from correlation function



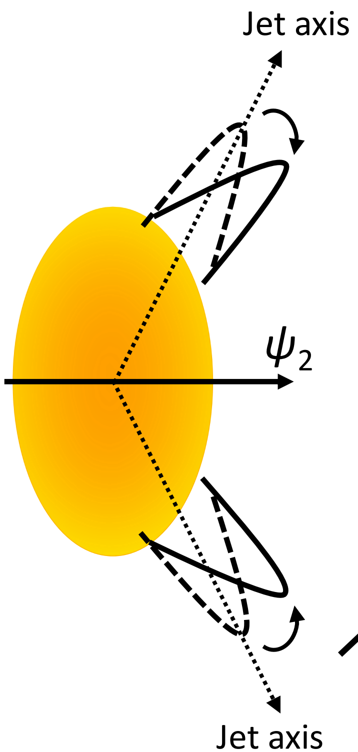
- Estimated background distribution by side-band method (green), fitting to side-band distribution (red), and Flow MC (blue)
  - Differences are assigned as a systematic uncertainty on background estimation

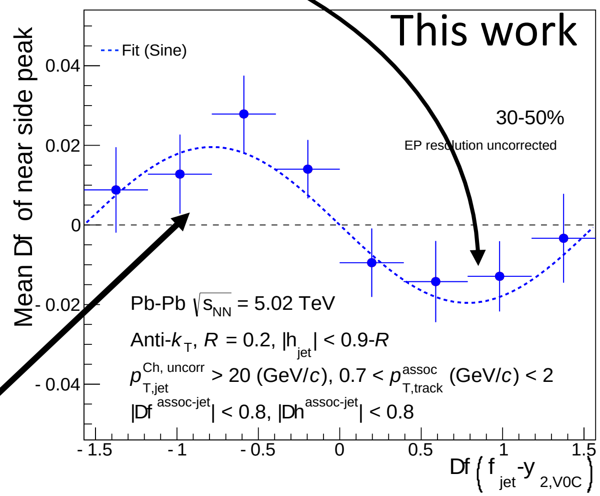


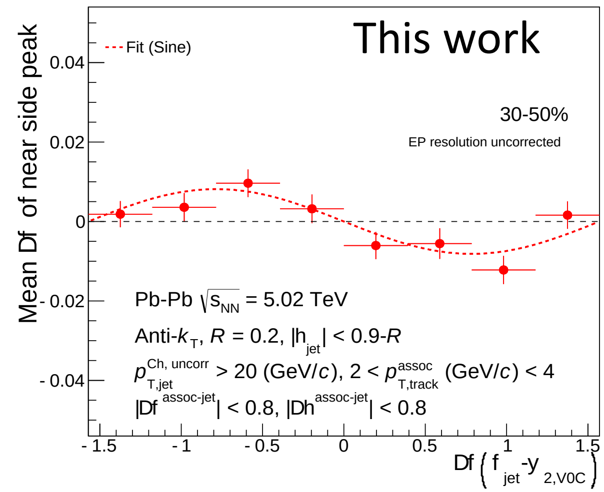
- Side-band background distribution was scaled based on the ratio of projection range (Scale factor: (|Δη|<0.8)/(1<|Δη|<1.5) → 1.6)

Near-side peak position shift

Fit by sine function with free parameter M: $f(x) = M \cdot \mathrm{Sin}(2x)$





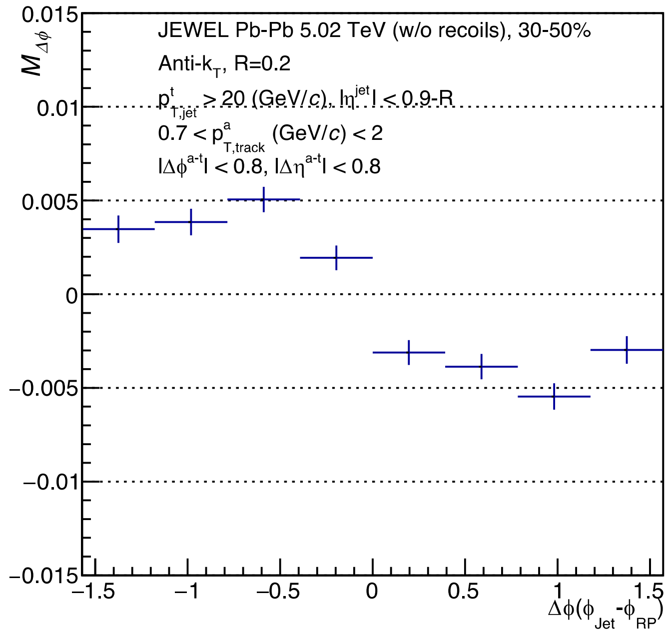


- **Mean position of the near-side peak tend to be biased towards in-plane**
- **The results suggests asymmetric jet modification w.r.t jet axis in mid-central collisions**
  - **Initial elliptical shape of media may leads to asymmetric jet modification**

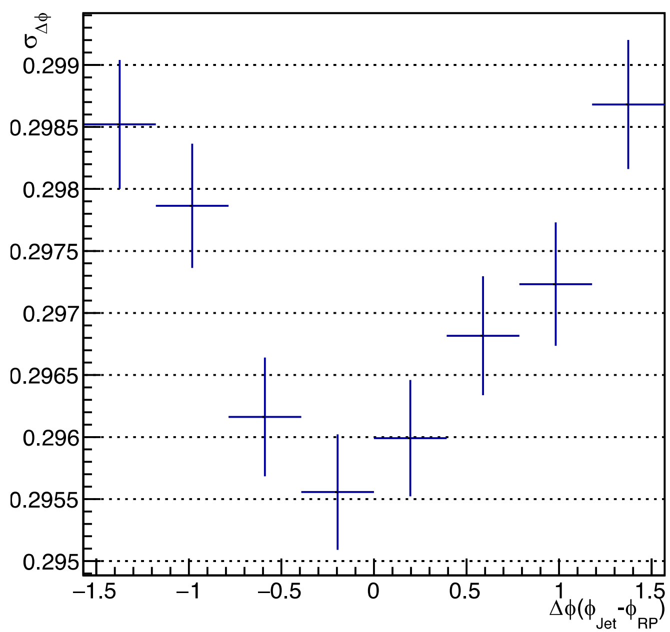
Prediction by JEWEL

- Mean and Sigma are calculated with mathematical expressions
- Any flow background subtraction is not applied for correlation functions

Mean



Width



- ➢ Qualitatively similar trends to the data
  - ➢ However, the effects seem to be much smaller than data though both results cannot be compared directory
    - ➢ How the results are changed if the recoil partons are taken into account?
    - ➢ What about other model predictions? (e.g. Energy loss + Hydro)

JEWEL: Eur.Phys.J. C74 (2014) 2762

Comparison to model prediction

- Measured results were compared to results of model prediction by JEWEL (Eur. Phys. J. C 74 (2014))

The ratio of near-side jet peak width in azimuth ($\Delta\varphi$)

p_{T}^{assoc} (GeV/c)	$0.7 < p_{\mathrm{T,track}}^{assoc} < 2$	$2 < p_{\mathrm{T,track}}^{assoc} < 4$
Data	1.22 ± 0.07 (stat.) $\pm^{0.05}_{0.11}$ (syst.)	1.00 ± 0.03 (stat.) $\pm^{0.04}_{0.09}$ (syst.)
JEWEL	1.02 ± 0.02 (stat.)	1.00 ± 0.03 (stat.)

The ratio of near-side jet peak width in eta ($\Delta\eta$)

p_{T}^{assoc} (GeV/c)	$0.7 < p_{\mathrm{T,track}}^{assoc} < 2$	$2 < p_{\mathrm{T,track}}^{assoc} < 4$
Data	1.07 ± 0.03 (stat.) $\pm$ 0.05 (syst.)	0.97 ± 0.02 (stat.) $\pm$ 0.02 (syst.)
JEWEL	0.98 ± 0.02 (stat.)	0.97 ± 0.02 (stat.)

The jet width difference between in-plane and out-of-plane was not reproduced by this model prediction

The amplitudes of fit function ($M \cdot \mathrm{Sin}(2\mathrm{x})$) to the peak position in azimuth

p_{T}^{assoc} (GeV/c)	$0.7 < p_{\mathrm{T,track}}^{assoc} < 2$	$2 < p_{\mathrm{T,track}}^{assoc} < 4$
Data	-0.02 ± 0.005 (stat.) $\pm$ 0.006 (syst.)	-0.008 ± 0.002 (stat.) $\pm$ 0.002 (syst.)
JEWEL	-0.004 ± 0.002 (stat.)	0.0006 ± 0.0007 (stat.)

The near-side peak shift effect is qualitatively reproduced only for low-p_{T} associates though it is tiny effect